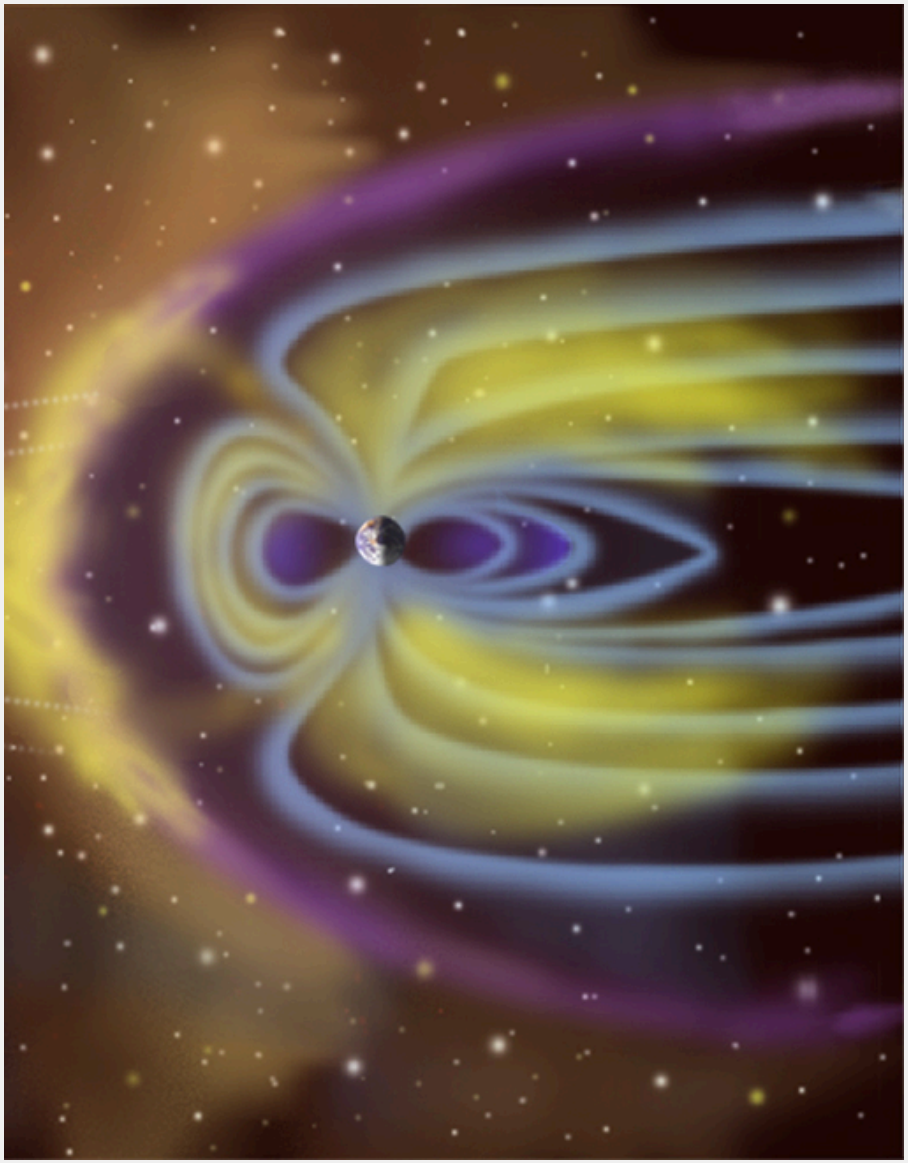
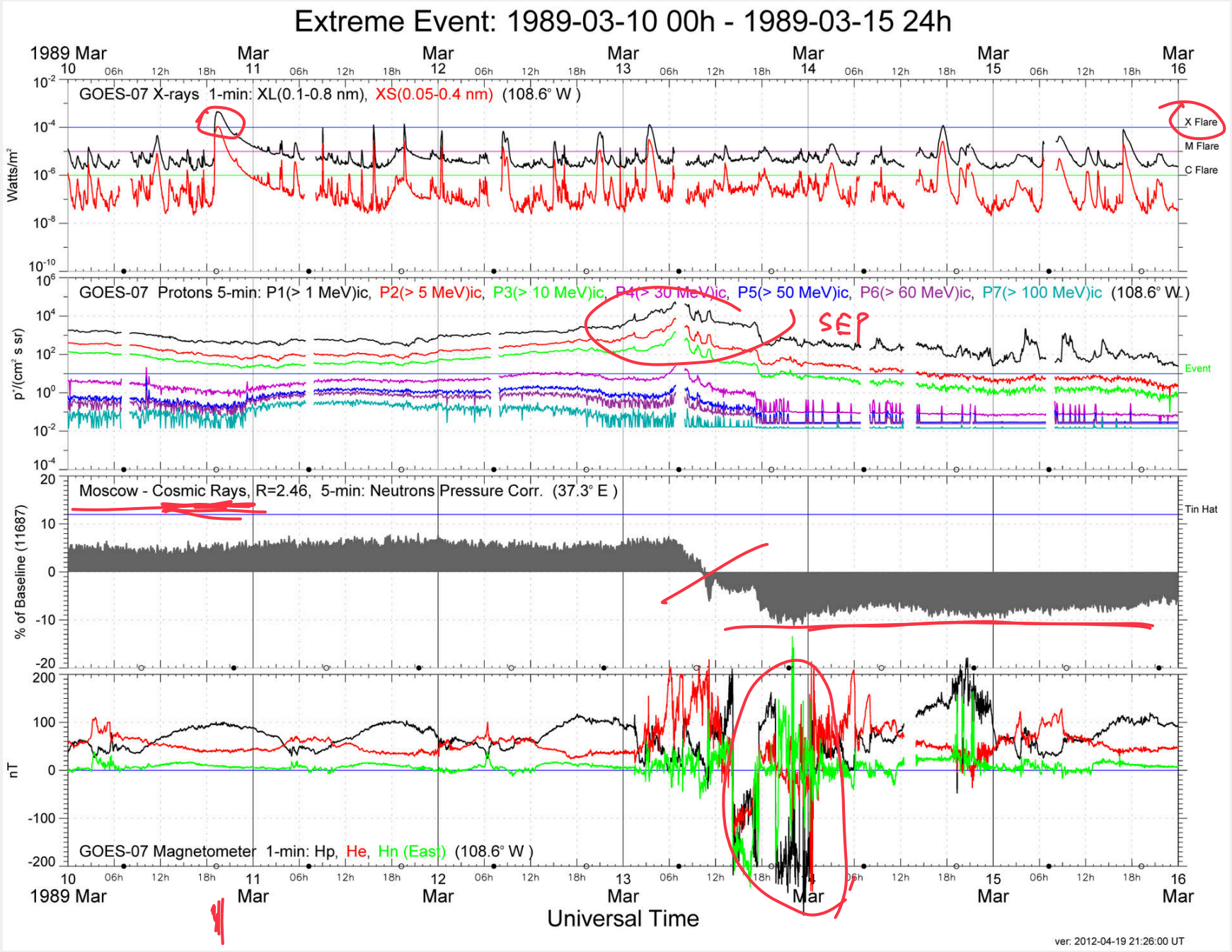
Storm Effects

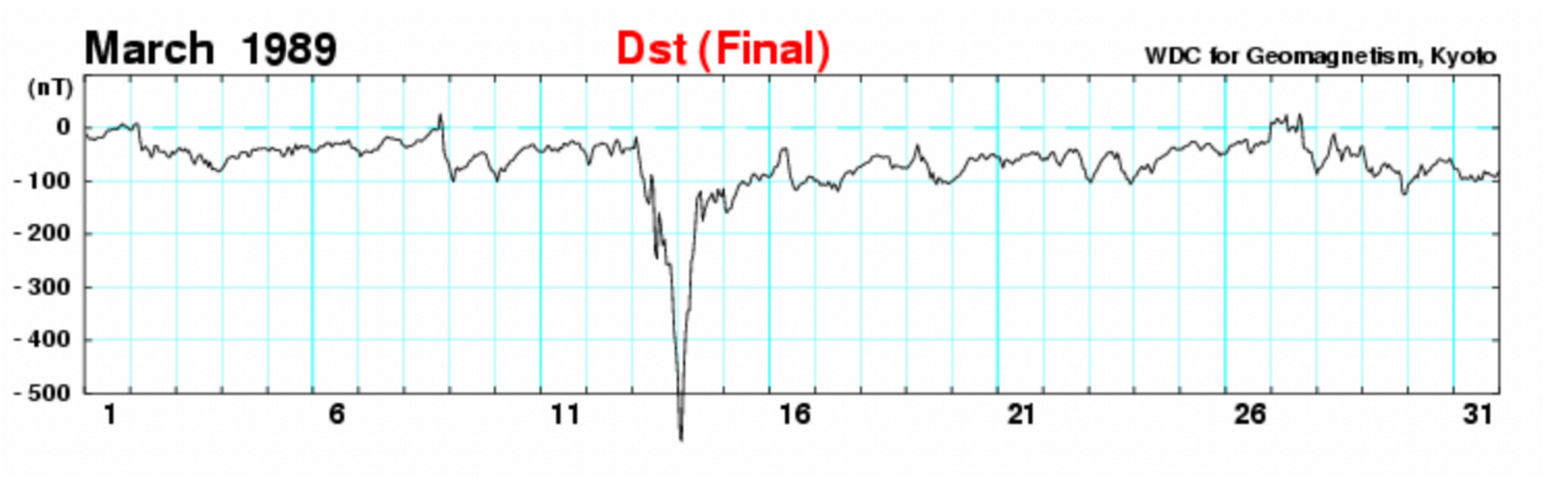
- **What happens to the space environment during a geomagnetic storm?**
  - Magnetosphere gets compressed; magnetopause moves inward
  - Reconnection and field line convection
  - Solar wind / CME plasma gets injected into magnetosphere
  - Ring current gets enhanced 10x or more
    - Ring current enhancement leads to reduction of B-field on the surface by 1–2% within hours; recovery over 1–2 days
  - Plasmasphere shrinks (being eroded): plasmapause moves inwards
  - Radiation belts get enhanced: 10-100x more high-energy electrons and protons
  - Auroras occur; larger storm, auroras reach to lower latitudes



Example: March 1989 Event

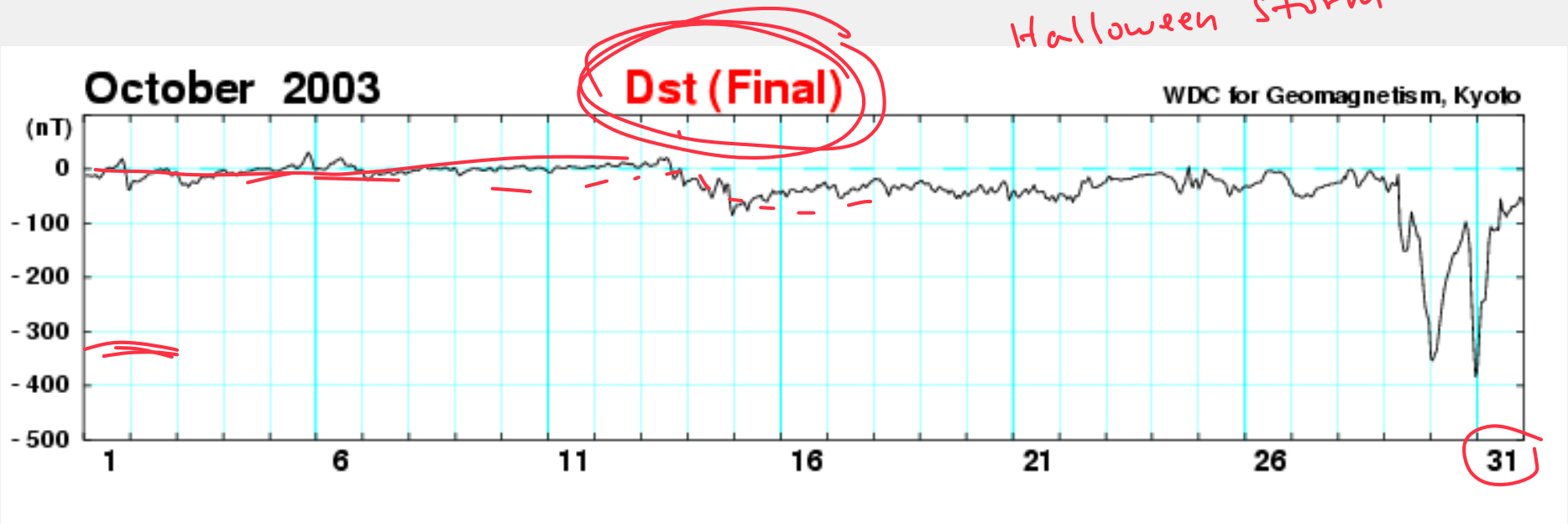


Dst index during March 1989. The severe geomagnetic storm of March 13–14 reached a minimum Dst of −589 nT and caused the collapse of the Hydro-Québec power grid, leaving approximately six million people without electricity for about nine hours.



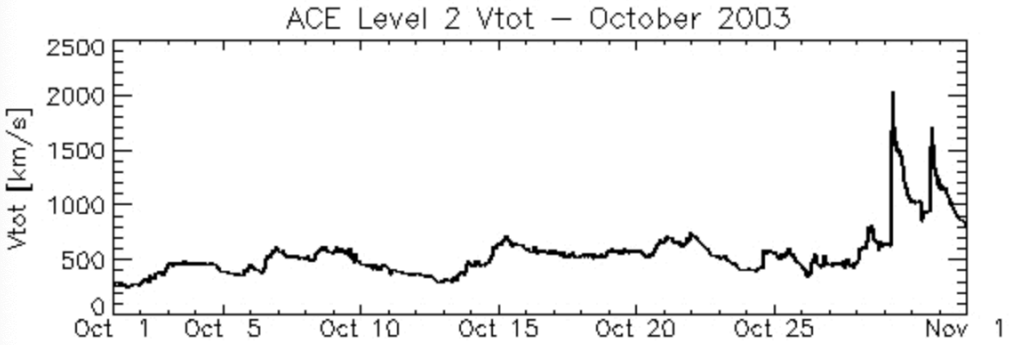
The strongest geomagnetic storm since 1957, as measured by the Dst index, which began in January 1957.

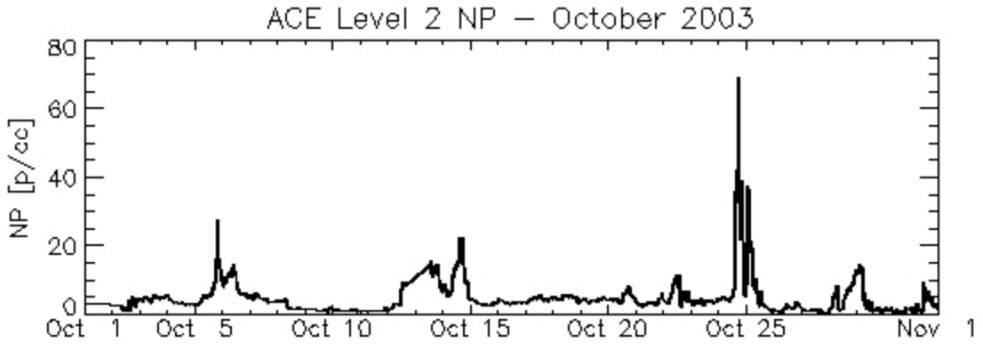
Geomagnetic Storm and Dst

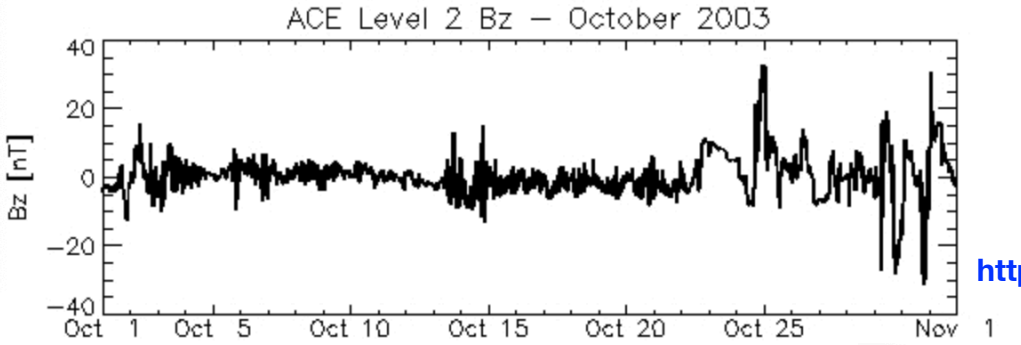


Continuous, near-real-time monitoring of the upstream solar wind became operational in January 1998, following the launch of the Advanced Composition Explorer (ACE) spacecraft in August 1997 and its deployment near the Sun–Earth L1 point.

Geomagnetic Storm and Dst Prediction

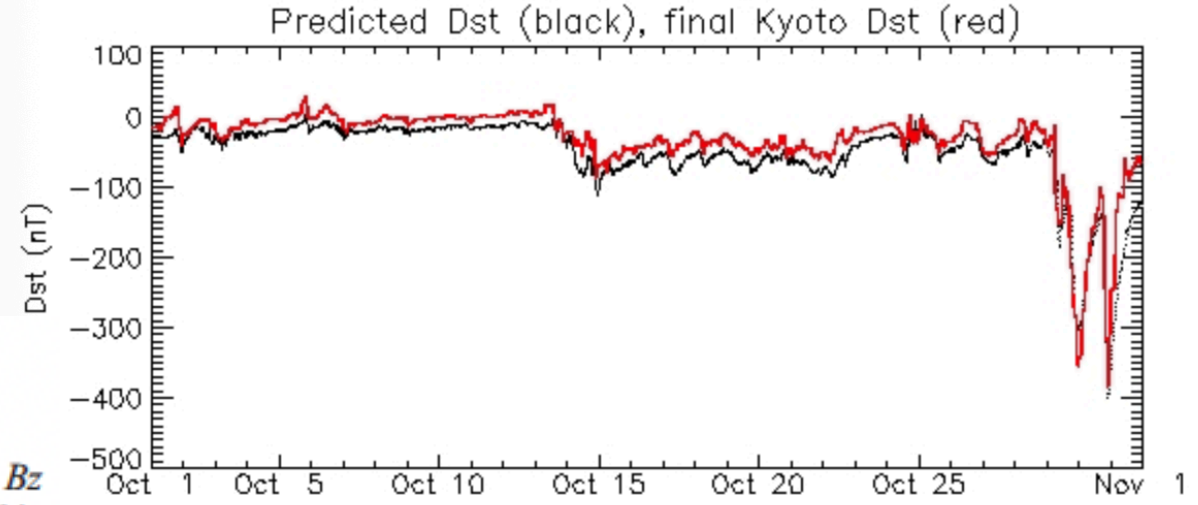






(Temerin and Li, JGR, 2002; 2006)

https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html

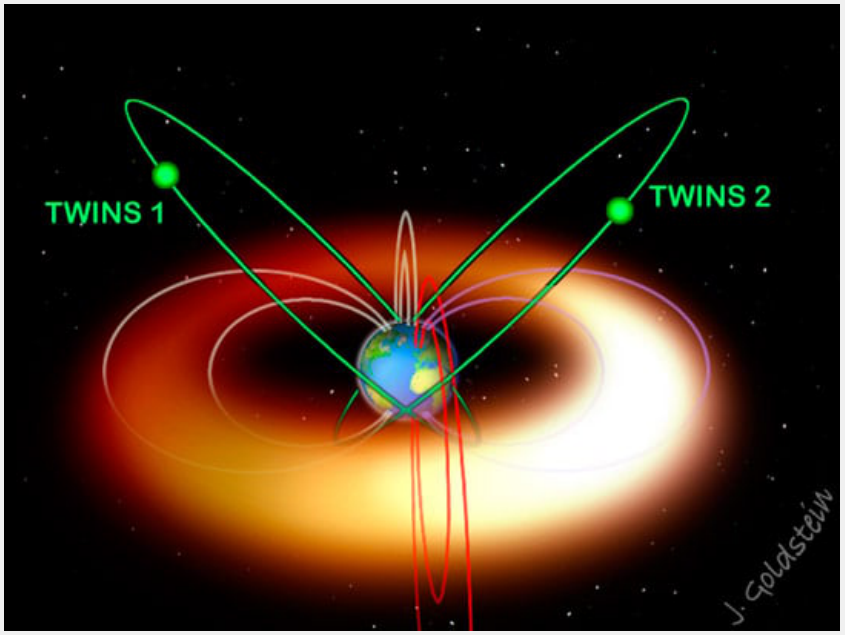


How is this possible?

$$Dst = dst1* + dst2* + dst3* + (pressure\ term)* + (direct\ Bz\ term)* + (offset\ term).$$

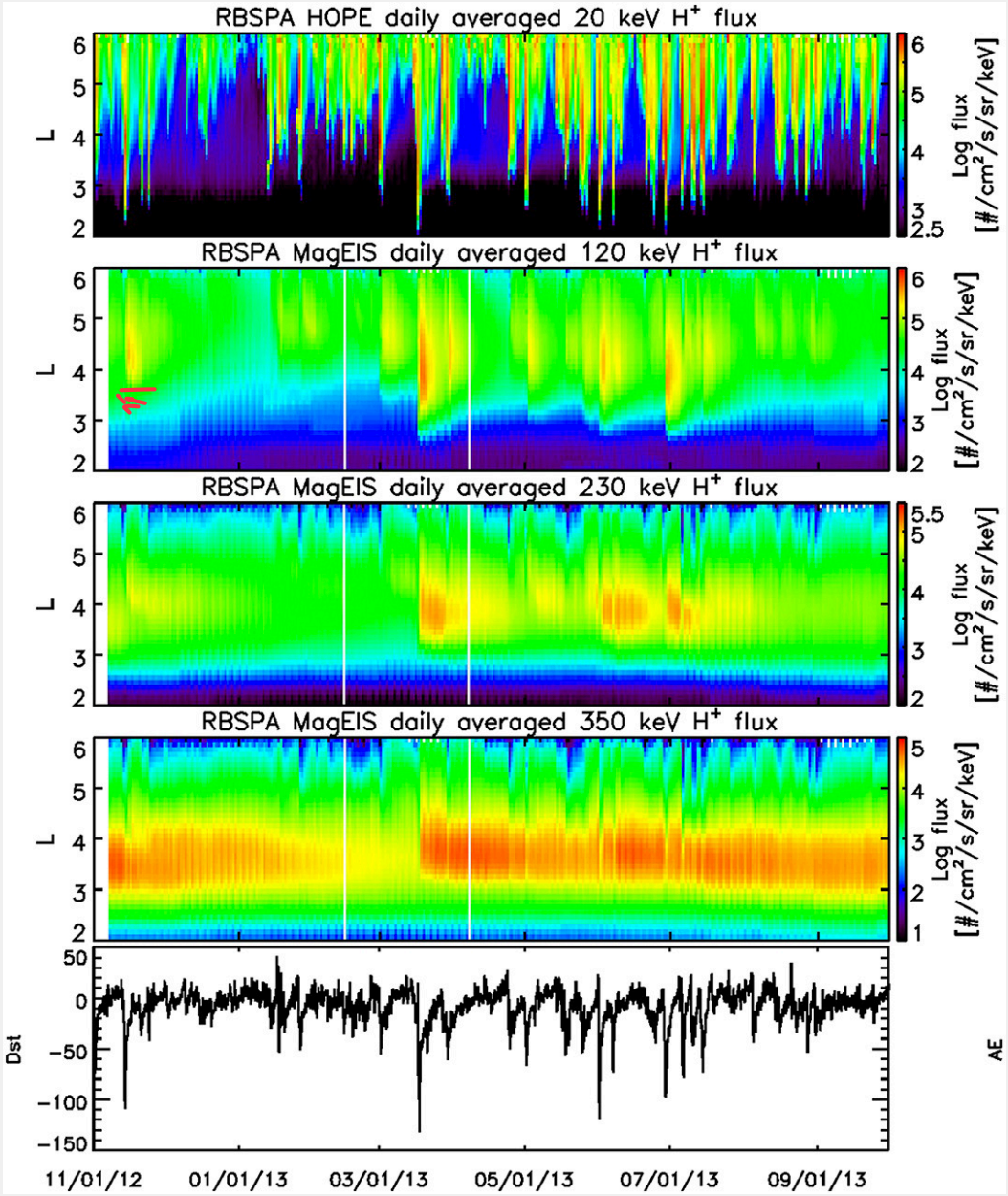
The terms *dst*1, *dst*2, *dst*3, (*pressure term*), and (*direct Bz term*) were in our previous model. The terms *dst*1*, *dst*2*, *dst*3*, (*pressure term*)*, (*direct Bz term*)* differ, in part, in

Ring Current Enhancement



Earth's Ring Current

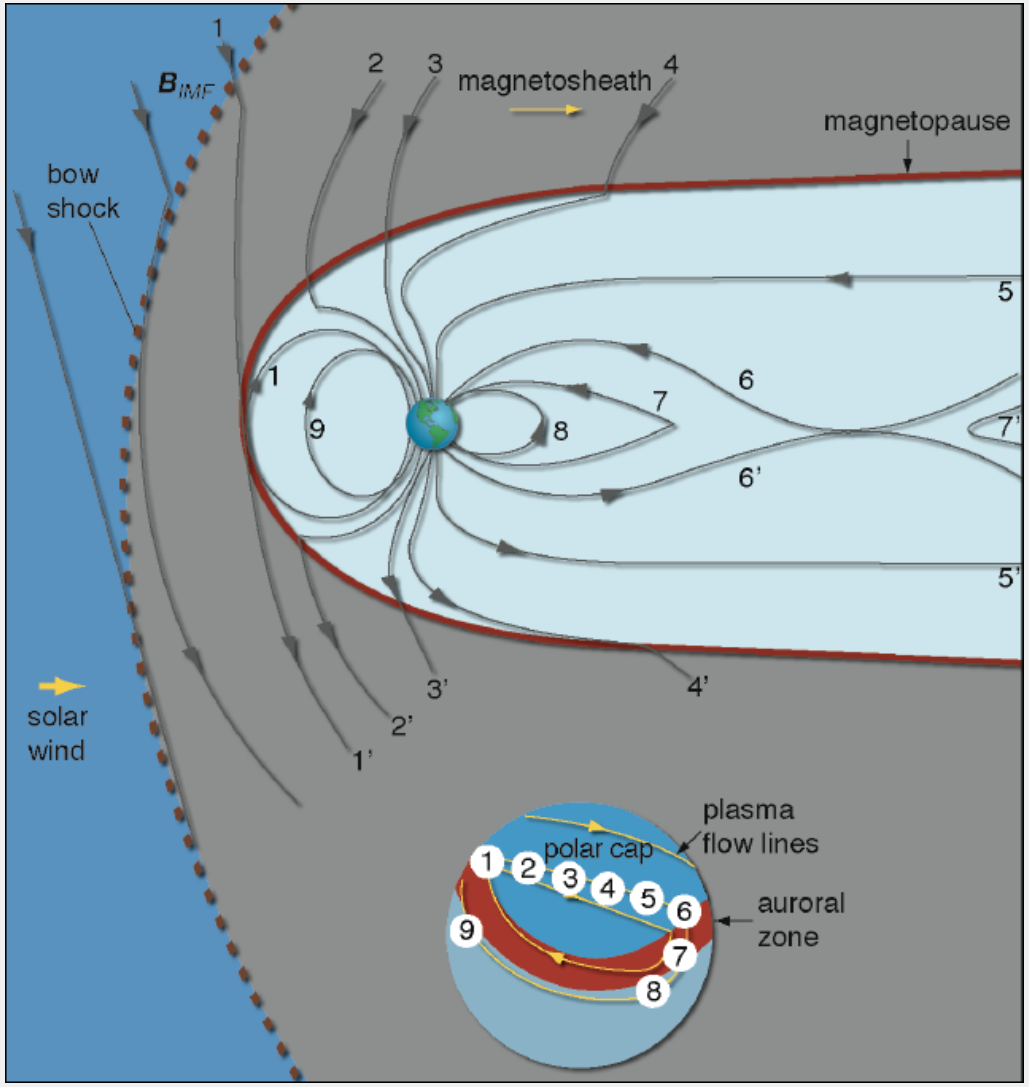
- **Second and Third panels** are the main ring current (protons)
- Enhancement during storm (red) is more than 10x higher than background (green)
  - Log scale
- Note correlation with Dst: good, but not perfect



(Zhao, Li, et al., JGR, 2015)

Geomagnetic Substorms

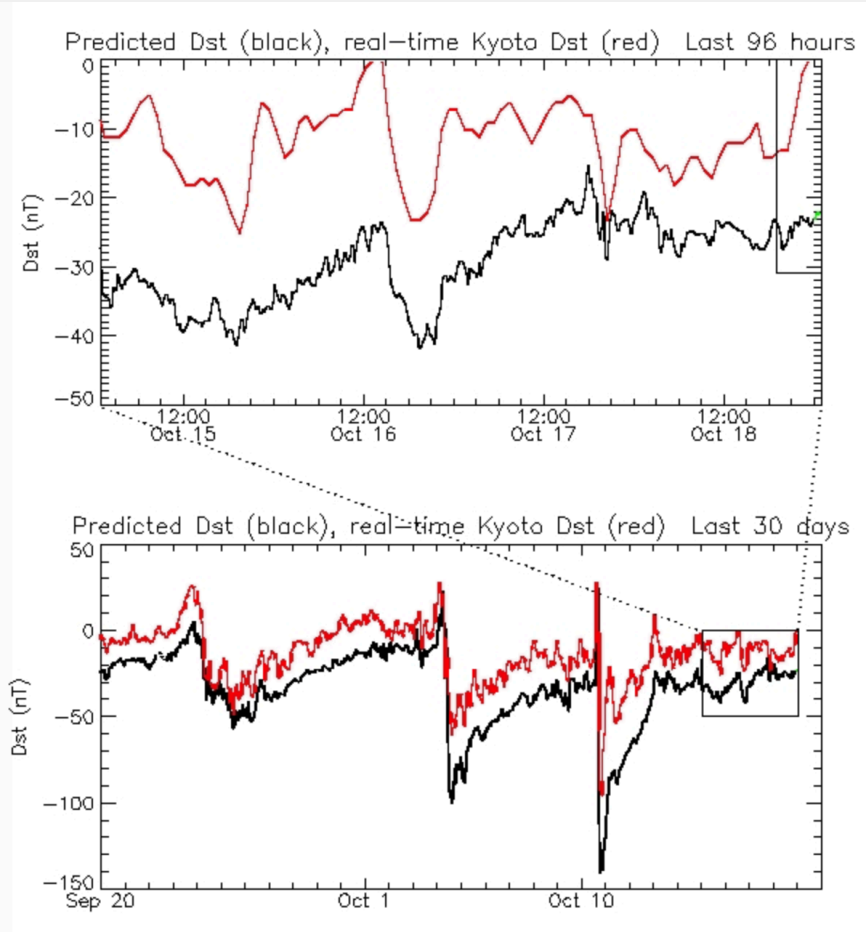
- **What is a geomagnetic substorm?**
  - driven by “quiescent", but unsteady, solar wind – small IMF fluctuating in direction
  - Occur 3–4 times per day
  - Energy stored in magnetotail plasma is released; leads to aurora
  - Measured as aurora and as B-field perturbations at **high latitude** (storms: at equator)
  - **AE index** (to be explained on next page) indicates the magnitude of the substorm



Geomagnetic Storms and Auroras

Geomagnetic Substorms and AE index Prediction

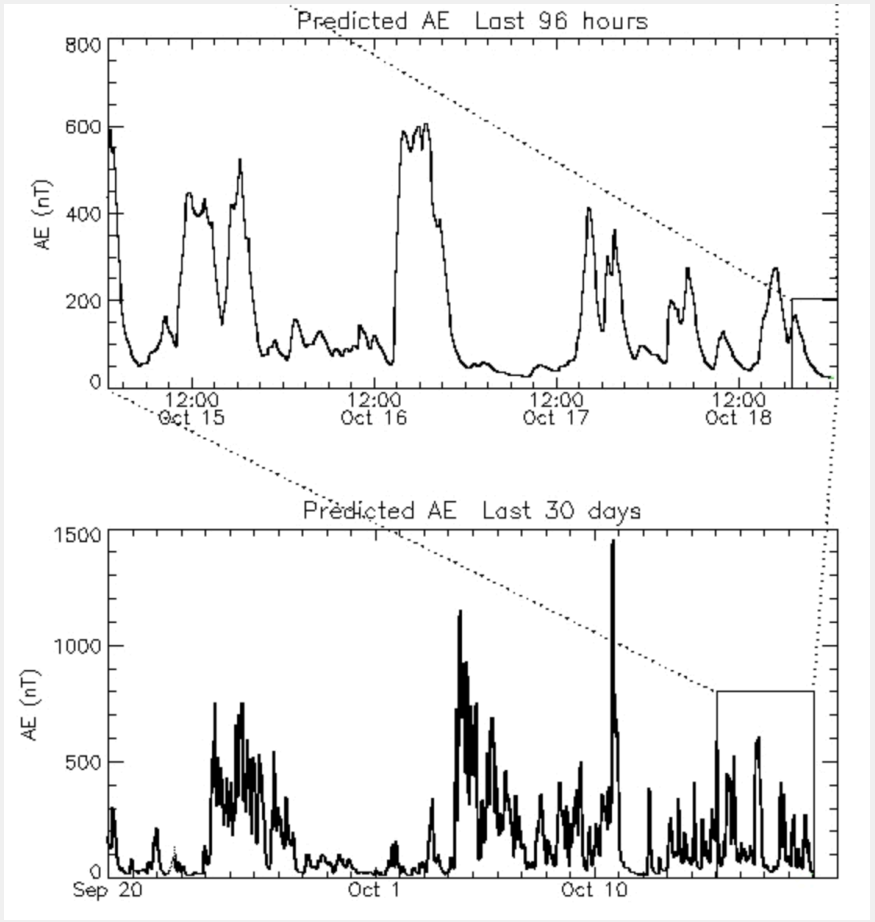
https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html



(Temerin and Li, JGR, 2002; 2006)

$$Dst = dst1* + dst2* + dst3* + (pressure\ term)* + (direct\ Bz\ term)* + (offset\ term).$$

The terms *dst*1, *dst*2, *dst*3, (*pressure term*), and (*direct Bz term*) were in our previous model. The terms *dst*1*, *dst*2*, *dst*3*, (*pressure term*)*, (*direct Bz term*)* differ, in part, in



(Li et al., JGR, 2007; Luo, B., X. Li et al., JGR, 2013)

The Auroral Electrojet (AE) index is derived from measurements by 12 ground-based magnetometers distributed around the Northern Hemisphere at geomagnetic latitudes of approximately 61°–70°. The AU and AL indices characterize the eastward and westward auroral electrojets, respectively, and their difference (AE = AU – AL) measures the overall auroral electrojet activity.

The AE index is also forecast in real time on our website.

Space Weather Forecast: spaceweather.com

spaceweather.com
News and information about the Sun-Earth environment

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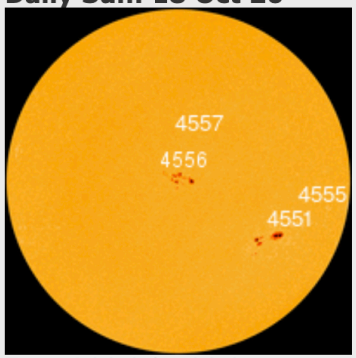
Current Conditions

Solar wind
speed: **404.6** km/sec
density: **2.0** protons/cm^3
more data: ACE, SOLAR-1
Updated: Today at 2356 UT

X-ray Solar Flares
6-hr max: **C2** 1917 UT Oct18
24-hr: **C3** 0644 UT Oct18
explanation | more data
Updated: Today at: 0001 UT

Daily Sun: 18 Oct 26



Expand: labels | no labels
Sunspot 4551 has a 'beta-gamma' magnetic field that harbors energy for M-class solar flares. However, fast-growing sunspot 4556 is of greater interest. It has doubled in size since yesterday and has a complex morphology typical of flaring sunspots. Credit: NASA/SDO

Sunspot number: 87
What is the sunspot number?
Updated 18 Oct 2026

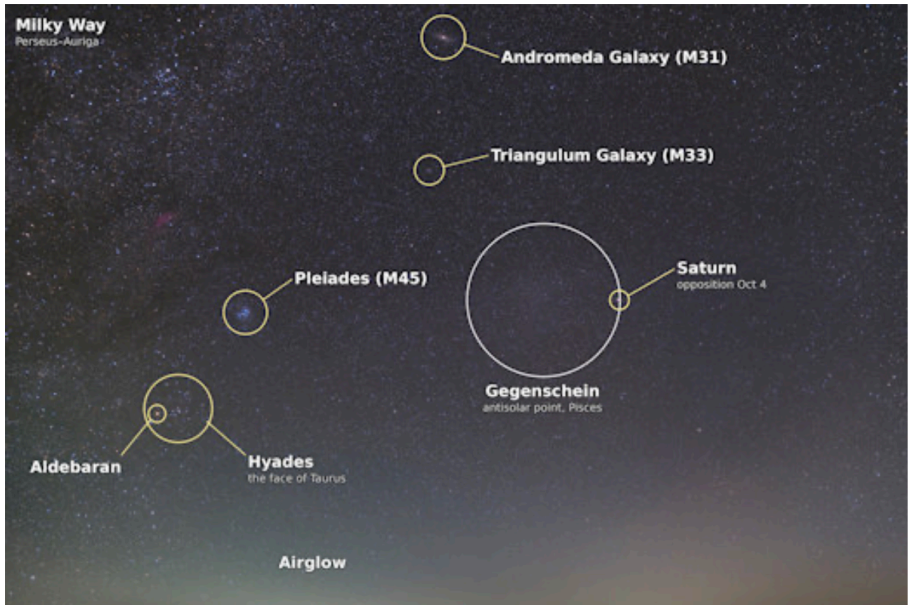
What's up in space

Monday, Oct. 19, 2026

AI writing is everywhere -- except here. This website is written by Dr. Tony Phillips, a carbon-based lifeform with 30 yrs of forecasting experience. Our guarantee: If you find a mistake, it was made by a real human being.

GEOMAGNETIC STORM WATCH (G1): A solar wind stream is approaching Earth. ETA: **Oct. 20th**. NOAA forecasters say that G1-class geomagnetic storms and high-latitude auroras are possible when it arrives. **Aurora alerts:** SMS Text.

THE SUN'S ELUSIVE COUNTERGLOW: It has been called "the skygazer's holy grail." In fact, most have never heard of it: Gegenshein is a faint patch of light in the night sky directly opposite the sun. Fins Eirexas photographed it on Oct. 15th from Mazaricos, Spain:



archives
October
19
2026
view

SATELLITE DRAG
today: **52.5** m/day ↑

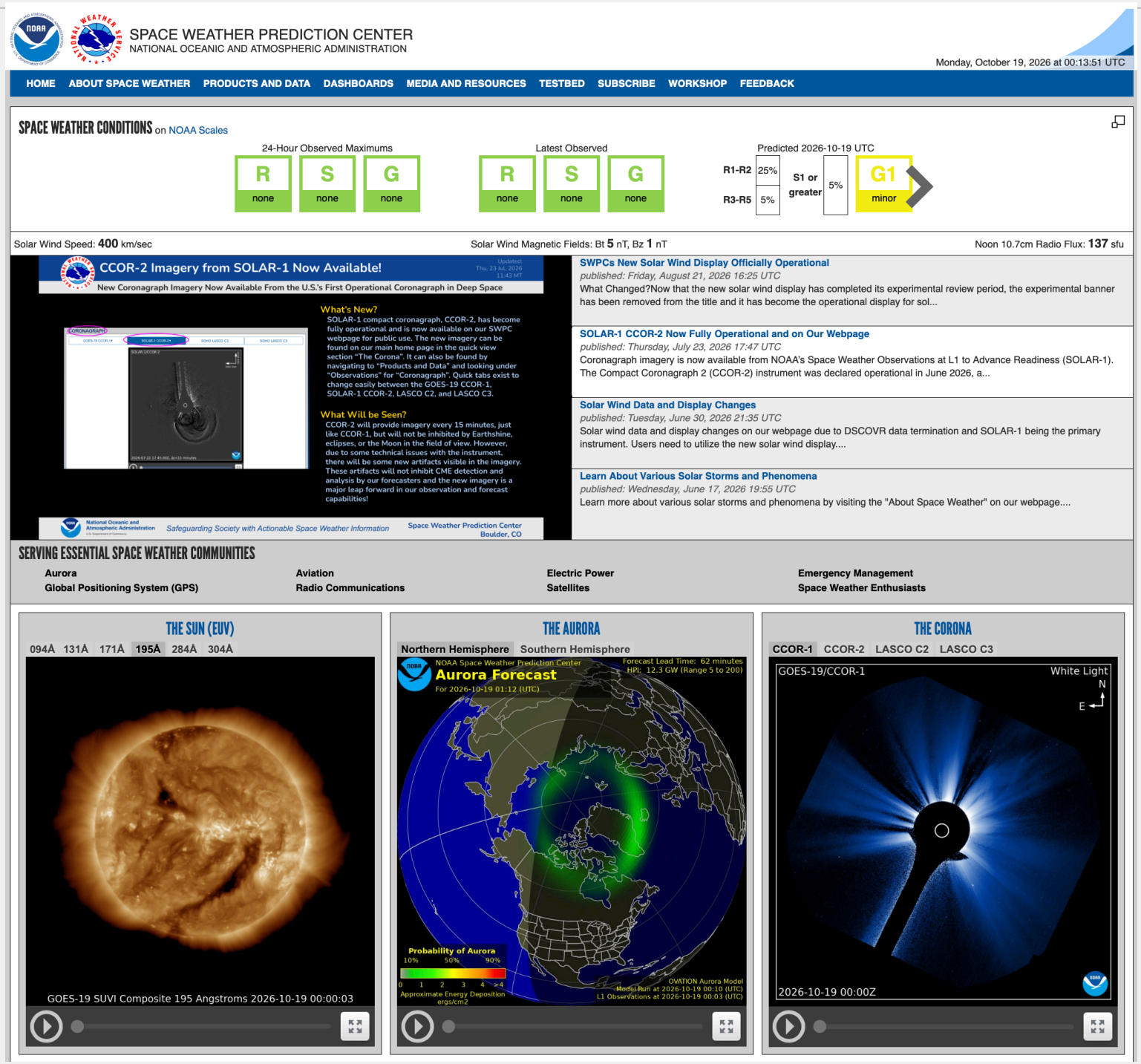
2026 max: **127.6** (Jan 20)
2026 min: **19.7** (Aug 10)

Explanation: This is how fast a typical Starlink at 480 km is sinking due to aerodynamic drag: more information.
Updated: 17 Oct 2026

Starlink Statistics
Total Launched: 13,015
In orbit: 11,154
Updated: 18 Oct 2026

Starlink Re-entries (2026)
January: 24
February: 54
March: 48
April: 51
May: 60
June: 35
July: 52
August: 32
September: 47
October: 29
Updated: 18 Oct 2026

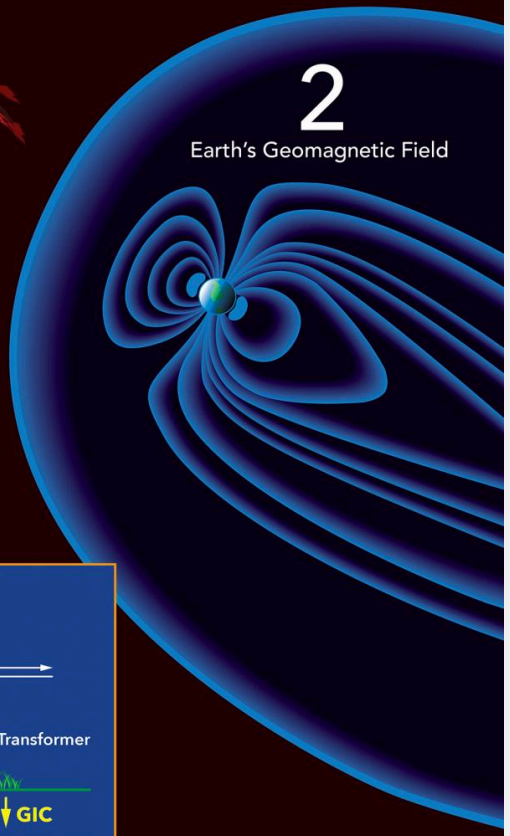
A bit more scientific and official: SWPC



National Aeronautics and Space Administration

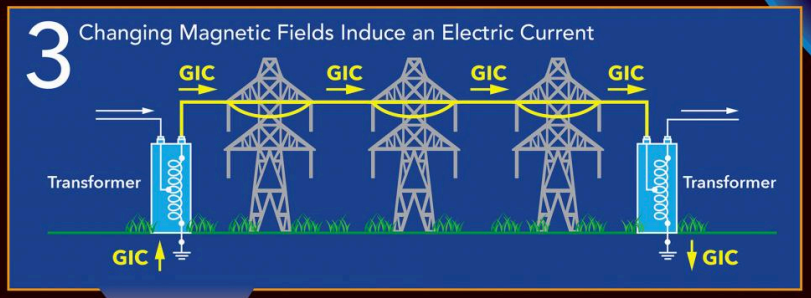
GEOMAGNETICALLY INDUCED CURRENTS

1
Coronal Mass Ejection (CME)



2
Earth's Geomagnetic Field

Geomagnetically Induced Currents (GICs) can result from geomagnetic storms—a type of space weather event in which Earth's magnetic field is rattled by incoming magnetic solar material. Most GICs are triggered by coronal mass ejections (1), or CMEs, which interact with the magnetic field around Earth (2) and cause it to rattle. The quick-changing magnetic fields create GICs through a process called electromagnetic induction (3). GICs can flow through railroad tracks, underground pipelines, and power grids. In extreme cases, they can cause blackouts.

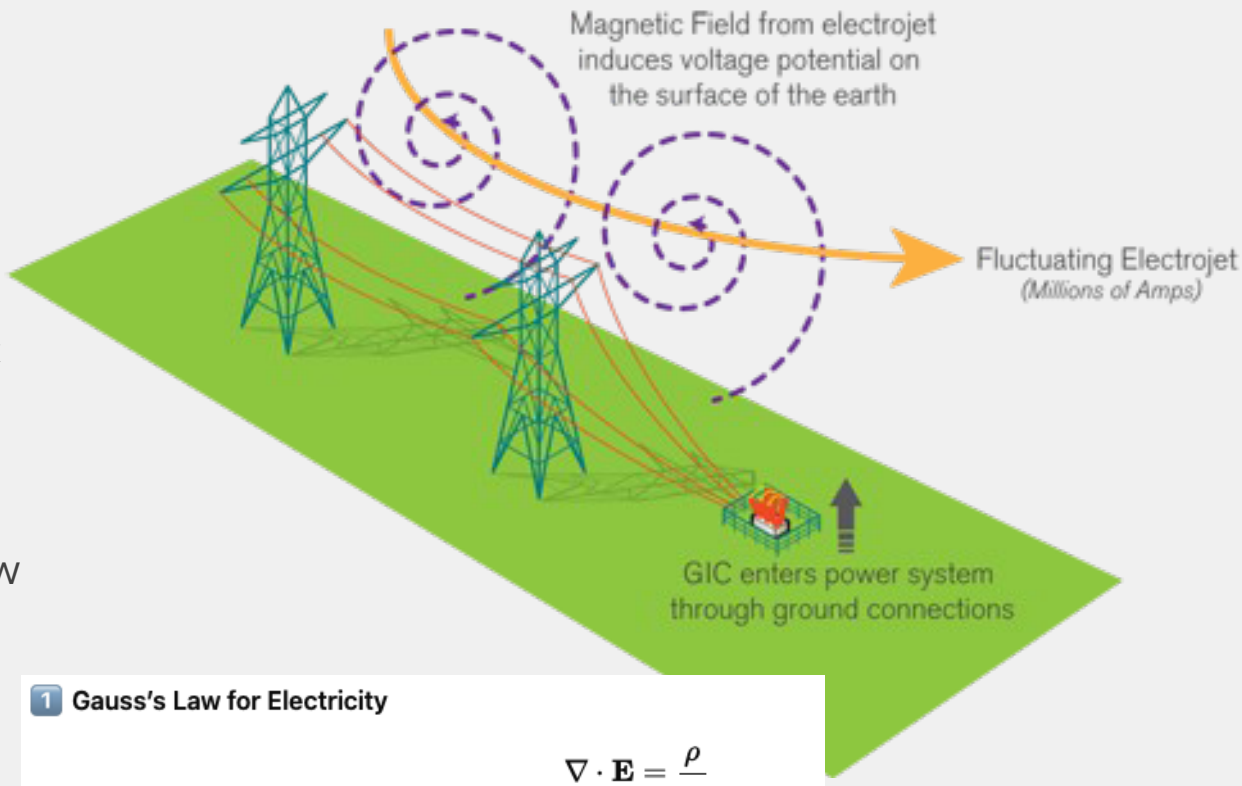


WHAT IS THE IMPACT?

Though widespread permanent damage to power systems is unlikely, extreme storms can cause blackouts over extended areas. That's why NASA and other federal agencies work with the power and insurance industries to develop plans and standards for dealing with GICs.

GIC mechanism

- Geomagnetic storms enhance currents in the ionosphere (e.g., the equatorial electrojet and auroral electrojet)
- This current produces a magnetic field
- magnetic field at surface induces currents in the earth (Faraday's law and Lenz's law)
  - Those currents would rather flow through long, conducting structures on the surface:
  - power /phone lines
  - railroads
  - pipelines



1 **Gauss's Law for Electricity**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

2 **Gauss's Law for Magnetism**

$$\nabla \cdot \mathbf{B} = 0$$

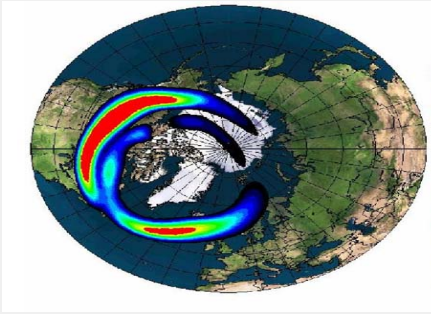
3 **Faraday's Law of Induction**

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

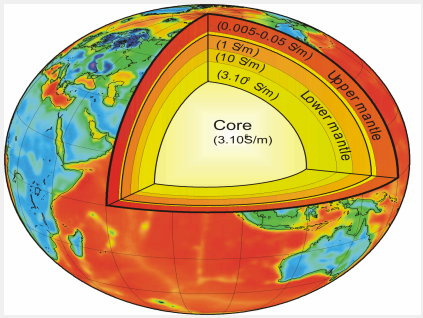
4 **Ampère–Maxwell Law**

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

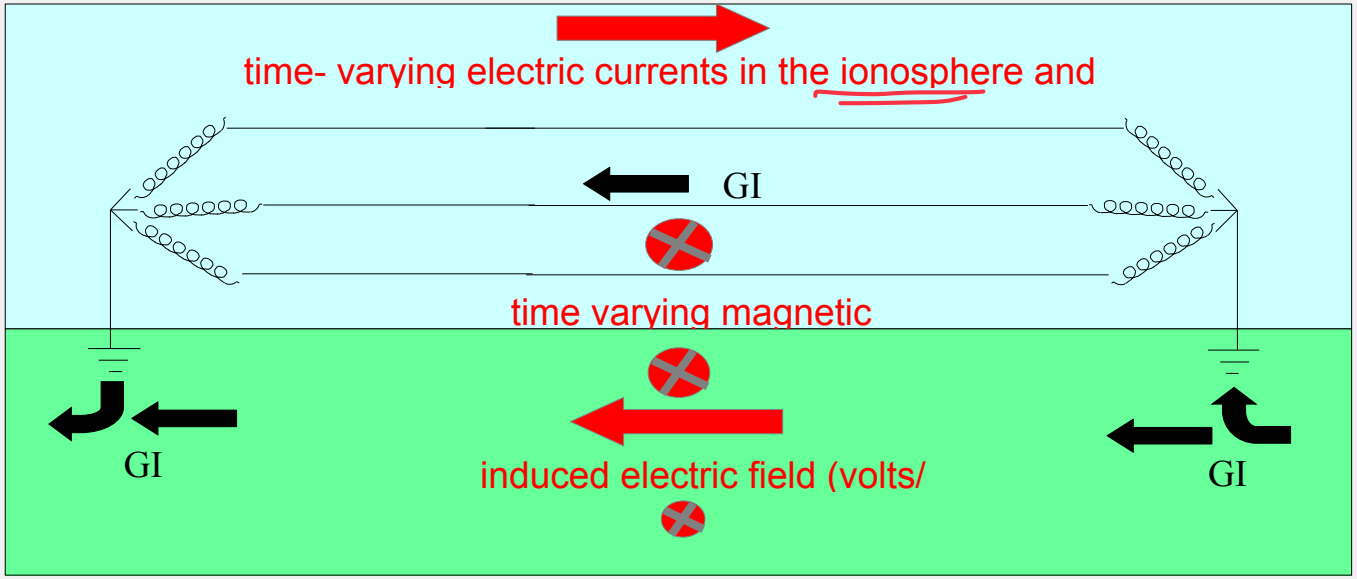
How does space weather cause grid problems?

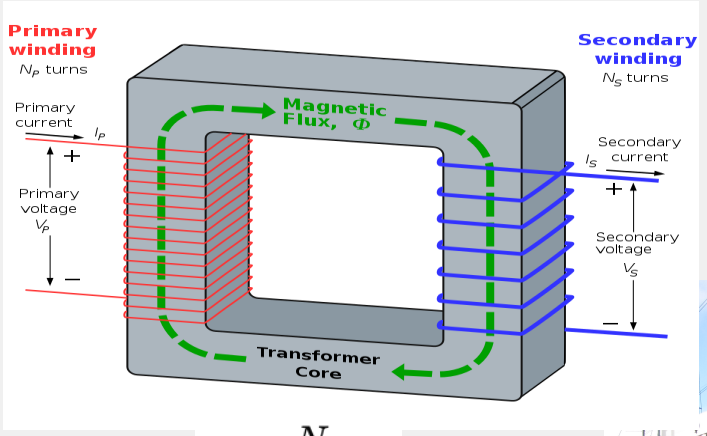


Electrical currents



Conducting Earth





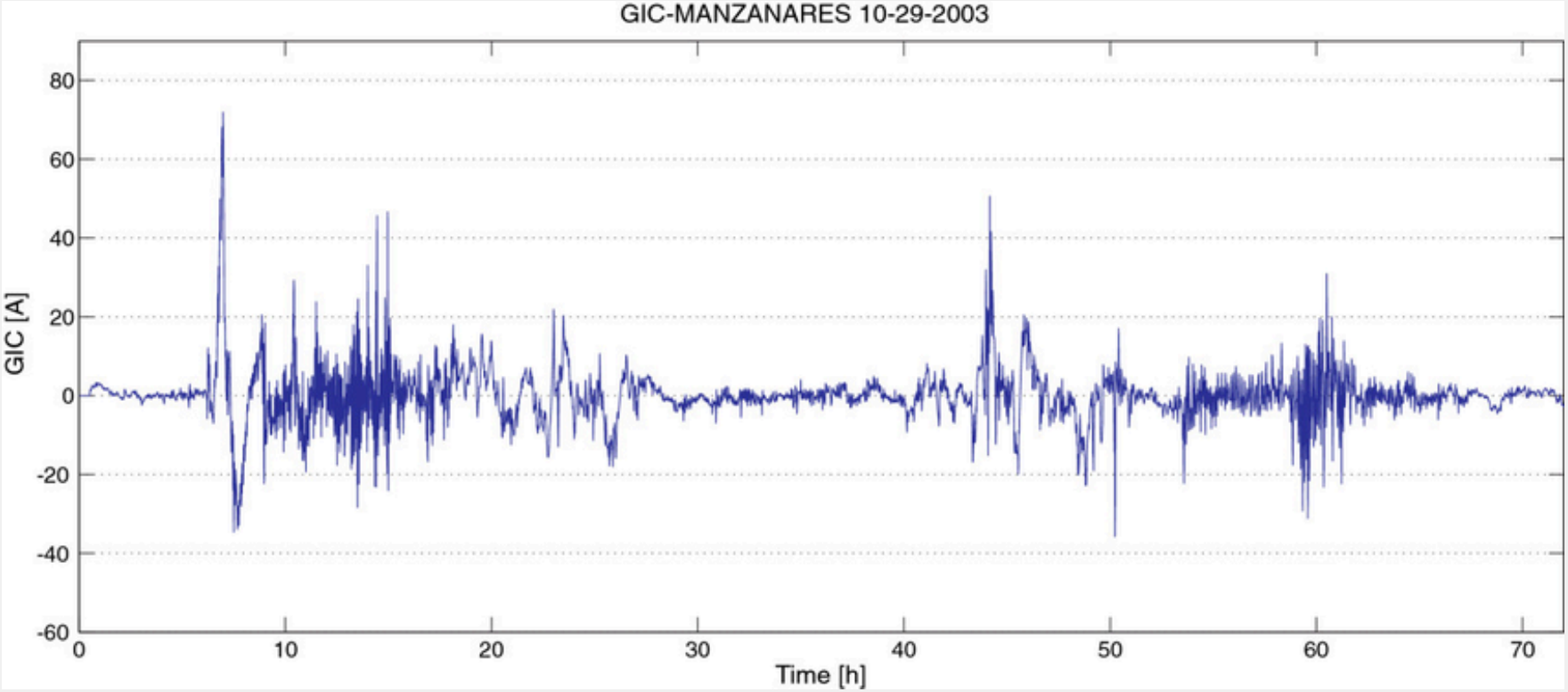
$$V_s = \frac{N_s}{N_p} V_p$$

Geomagnetically induced currents (GIC) cause

- Half-cycle saturation of transformers, voltage harmonics, overheating, increased reactive power demand, and/or drop in system voltage.
- Leading to **transformer burn-out** (in rare big storms) or **shortened transformer lifetimes** (due to many smaller storms).

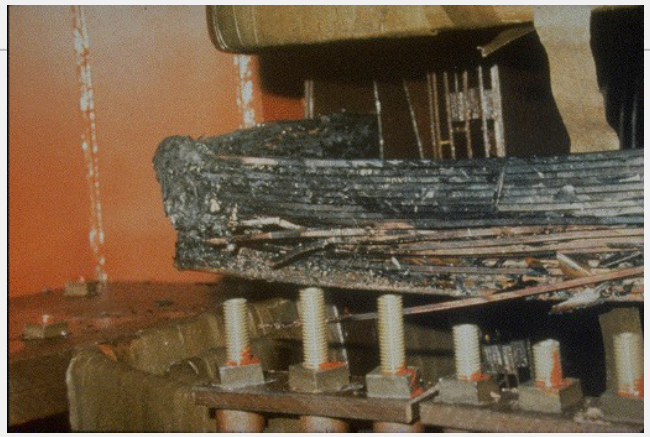
GIC magnitudes

- GICs can be up to tens of amperes during large storms
  - Long-distance power lines operate at very high voltage… to keep the current low (remember P = IV). Current spikes of tens of amps can overwhelm transformers



GIC effects

- **Power lines, transformers, substations:**
  - Voltage can be reduced, causing power system to shut down: blackout
  - Unlikely to cause transformer damage: requires sustained currents
- **Railroads:**
  - Switches operate automatically: switch opens if metal wheels are present, allowing current to flow (cool!)
- **Pipelines:**
  - Pipelines are held at negative potential to reduce corrosion (preventing electron loss)
  - Current flowing in pipelines could create false alarms, or reduce lifetime



Space Weather®

Research Article | Open Access

Modeling the Impact of Geomagnetically Induced Currents on Electrified Railway Signaling Systems in the United Kingdom

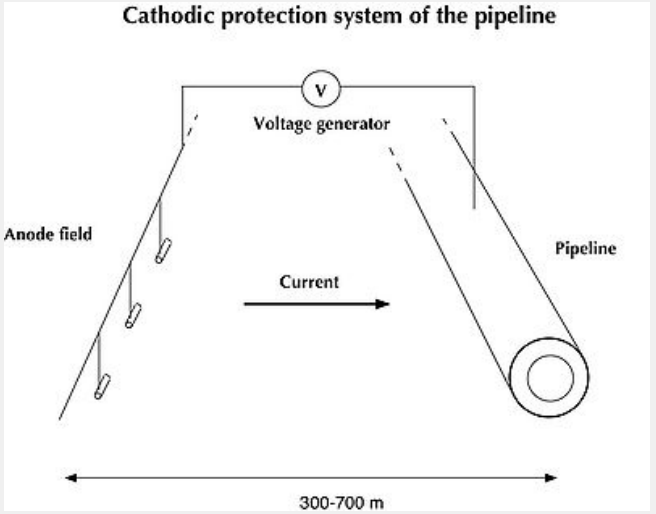
C. J. Patterson, J. A. Wild, D. H. Boteler

First published: 16 March 2023 | https://doi.org/10.1029/2022SW003385 | Citations: 5

CU Full Text

SECTIONS PDF TOOLS SHARE

Abstract



Effects of railways

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Rail industry urged to consider safety risks of space weather

Date: December 11, 2023

Source: Lancaster University

Summary: Train accidents could be caused by solar storms switching signalling from red to green according to new research examining the impact of space weather. Solar storms can trigger powerful magnetic disturbances on Earth, creating geomagnetically induced currents which could potentially interfere with electricity transmission and distribution grids. A study of two rail lines showed that more serious 'wrong side' failures (red to green) could occur with a weaker solar storm than for 'right side' failures, posing a serious risk which the industry needs to take on board.

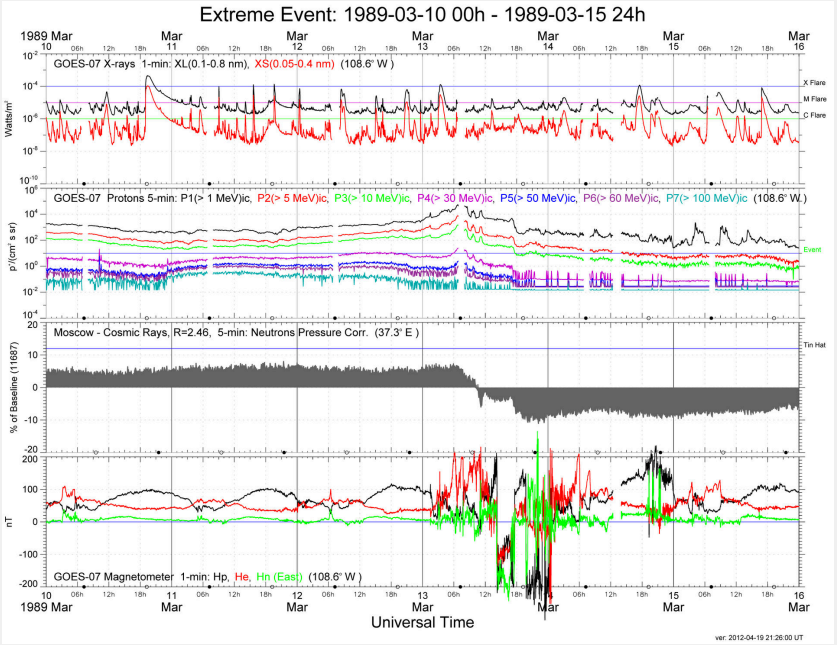
Share:

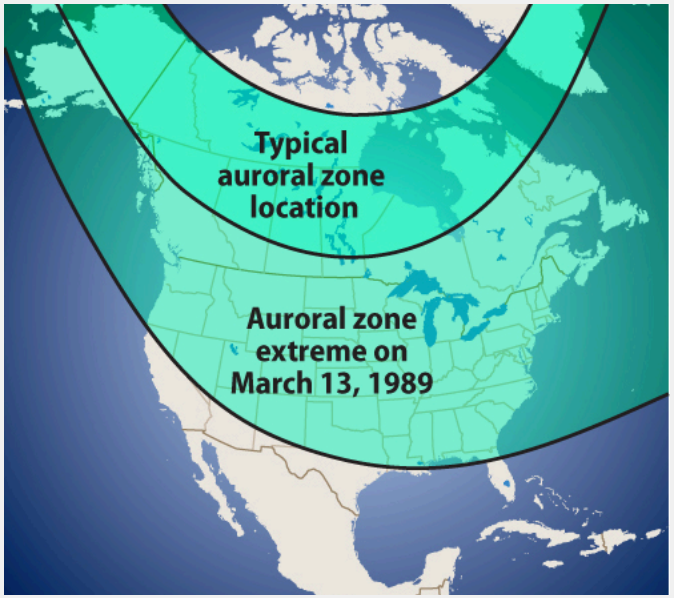
Breaking *this hour*

- Wearable Cameras Allow AI to Detect Med Errors
- Giant Meteorite Had Silver Lining for Life
- El Nino Oscillation 250 Million Years Old
- Role of Climate Change in Drought Intensity
- Plant Guard Cells Count Environmental Stimuli
- Plant CO2 Uptake Up by Nearly a Third Globally
- Rare Fossils of Extinct Elephants: Butchery
- New Insights Into Air Pollution Formation
- Scientists Build Modules for Synthetic Cell
- Life-Saving Spongelike 'Bandage'

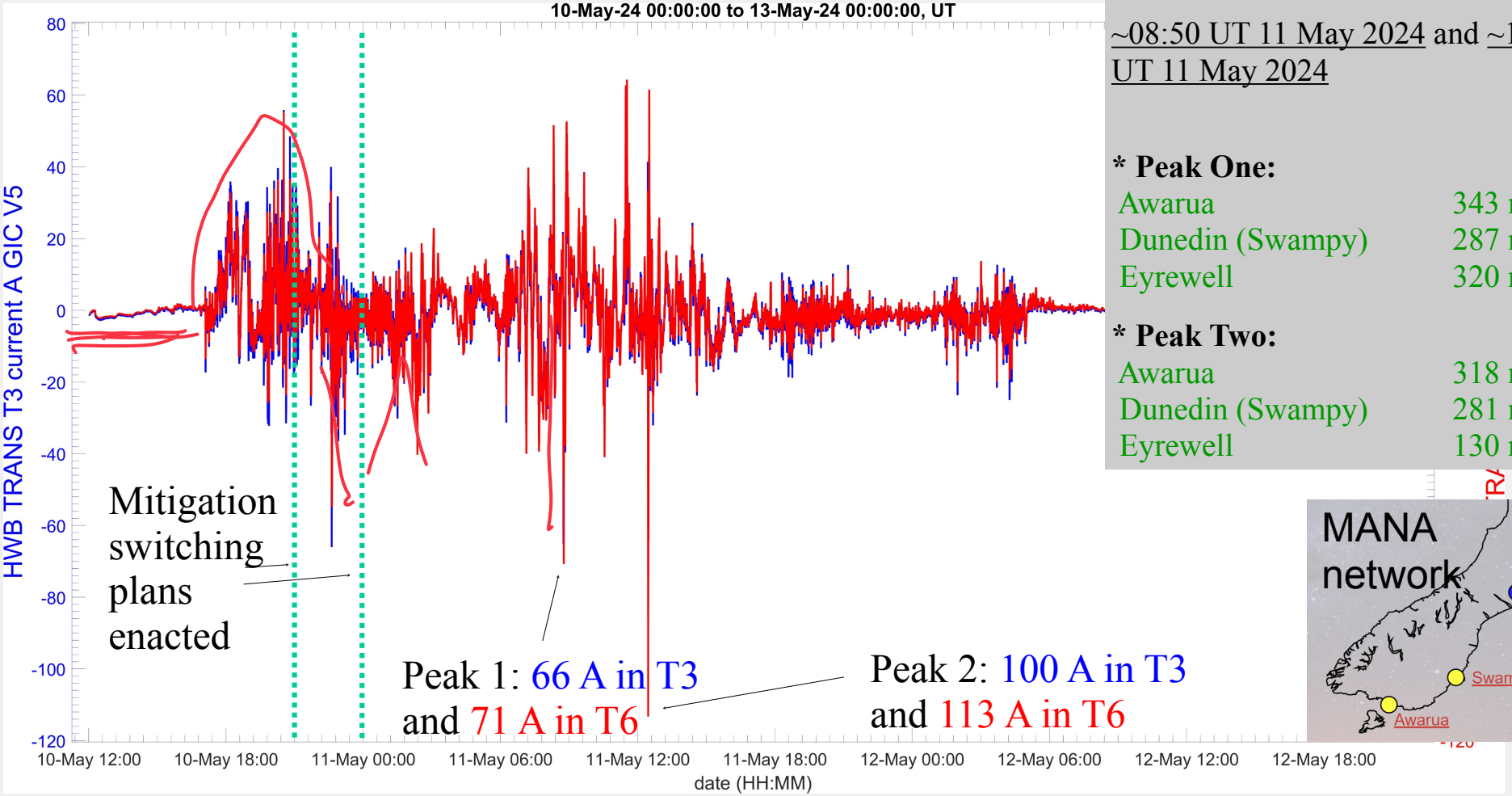
March 1989 Event: Quebec Blackout

- CME ejected from the sun on March 10, 1989: X-ray flare
- SEP event shortly after
- CME arrived three days later; massive magnetic field perturbations
- Aurora as far south as Texas!
- Quebec sits on top of poorly-conducting igneous rock: “The Canadian Shield”
- Currents flowed through the 735 kV power lines and tripped breakers; power outage for six millions people for nine hours
  - In Quebec
  - In March
  - Where it is cold and dark
- 200 other power grid “hickups” occurred across US





GIC observations from NZ during the Storm in May 2024



There are two significant peaks in magnetic field dH/dt,

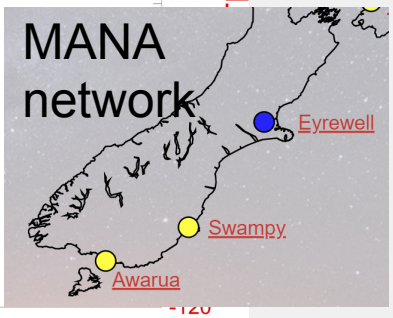
~08:50 UT 11 May 2024 and ~12:29 UT 11 May 2024

*** Peak One:**

Awarua	343 nT/min
Dunedin (Swampy)	287 nT/min
Eyrewell	320 nT/min

*** Peak Two:**

Awarua	318 nT/min
Dunedin (Swampy)	281 nT/min
Eyrewell	130 nT/min



The Carrington Event and other major events

- Two big storms in 2024 had Dst ~ −400 nT
- March 13, 1989 storm had Dst = −600 nT
- September 1, 1859 storm had Dst = −1600 nT: **The Carrington Event** (named for astronomer who observed the flare)
  - Aurora as far south as the Caribbean
  - Telegraph systems all over US and Europe failed; operators shocked (literally, and I suppose figuratively)

Modeling of 1–2 September 1859 super magnetic storm

Xinlin Li [a,*,1], M. Temerin [b], B.T. Tsurutani [c], S. Alex [d]

ScienceDirect

Advances in Space Research 38 (2006) 273–279

- An event of similar magnitude erupted from the sun in 2012, but missed the Earth by 9 days (out of a 365 day orbit, that's close!

A major solar eruptive event in July 2012: Defining extreme space weather scenarios

D. N. Baker,[1] X. Li,[1] A. Pulkkinen,[2] C. M. Ngwira,[2,3] M. L. Mays,[2,3] A. B. Galvin,[4] and K. D. C. Simunac[4]

SPACE WEATHER, VOL. 11, 585–591, doi:10.1002/swe.20097, 2013

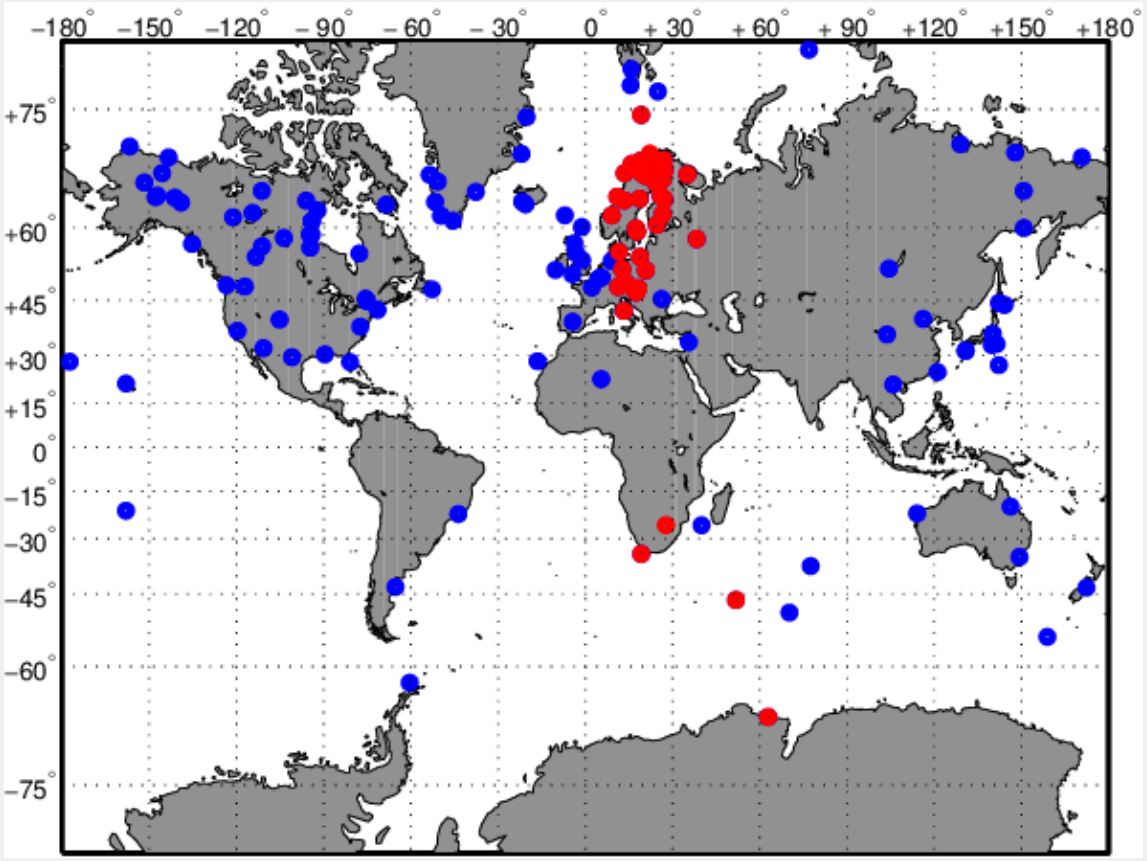
- A similar event today would cause **$1-3 trillion** in damages

University of Colorado **Boulder**

The Magnetosphere: Missions and Movies

Magnetic Field data… for the models

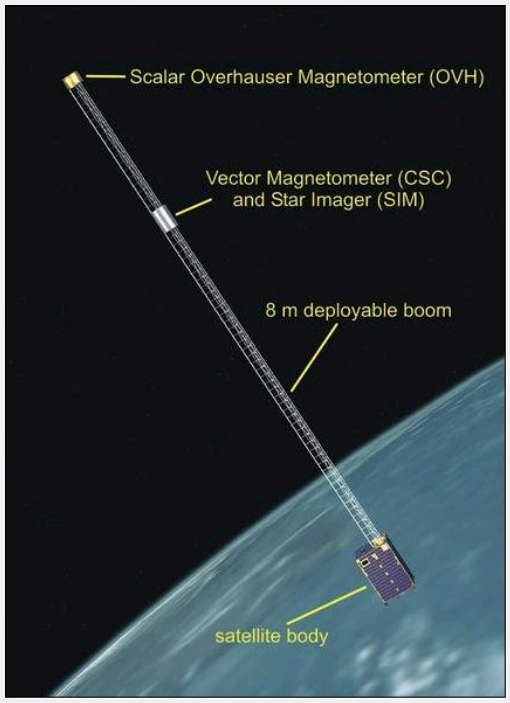
- The global magnetic field models (WMM and IGRF) primarily rely on ground magnetometers
- Major holes in coverage: Great in North America, Japan, and Europe, not so much everywhere else
- Use spacecraft data to fill those holes and map higher-order structure

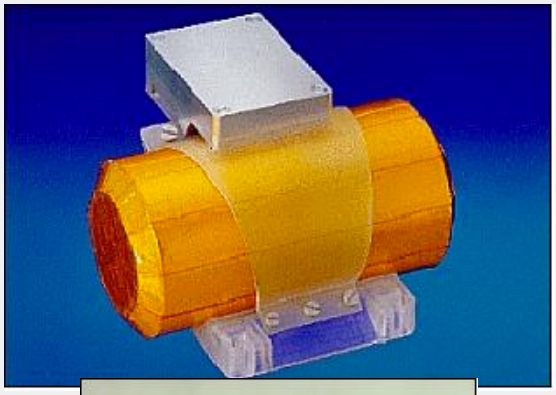


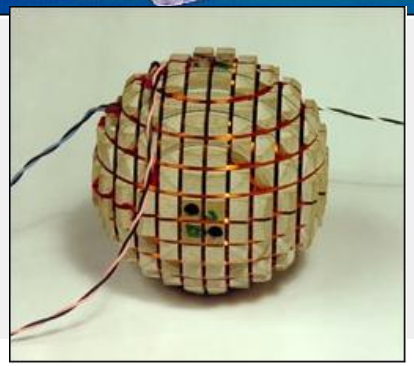
- 1979 – 1980: MAGSAT (NASA)
- 1990 – 2014: Ørsted
- 2000 – 2010: CHAMP
- 2013 – 2025: SWARM
- 2025 and beyond: COSMO (6U CubeSat, PI: Prof. Marshall), in commissing phase

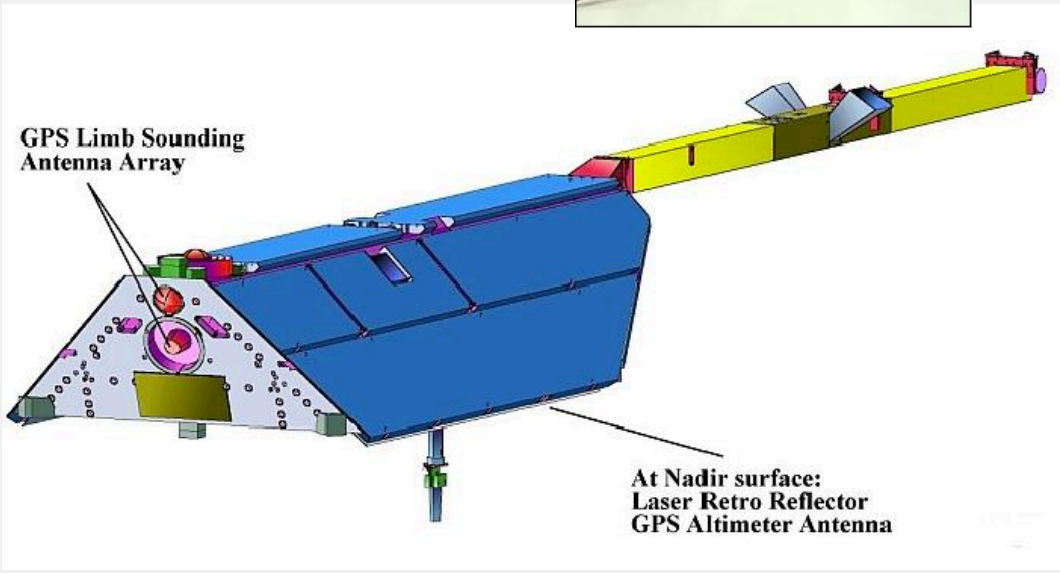
Ørsted and CHAMP

- **Ørsted: Danish micro-satellite**
  - ~60 kg spacecraft
  - Magnetometer on 8-meter boom
  - 655/857 km altitude
- **CHAMP: ESA mission with multiple science objectives**
  - 522 kg spacecraft
  - Magnetometers on 4 m boom; boom points in ram direction
  - 418/474 km altitude: 5 year nominal lifetime



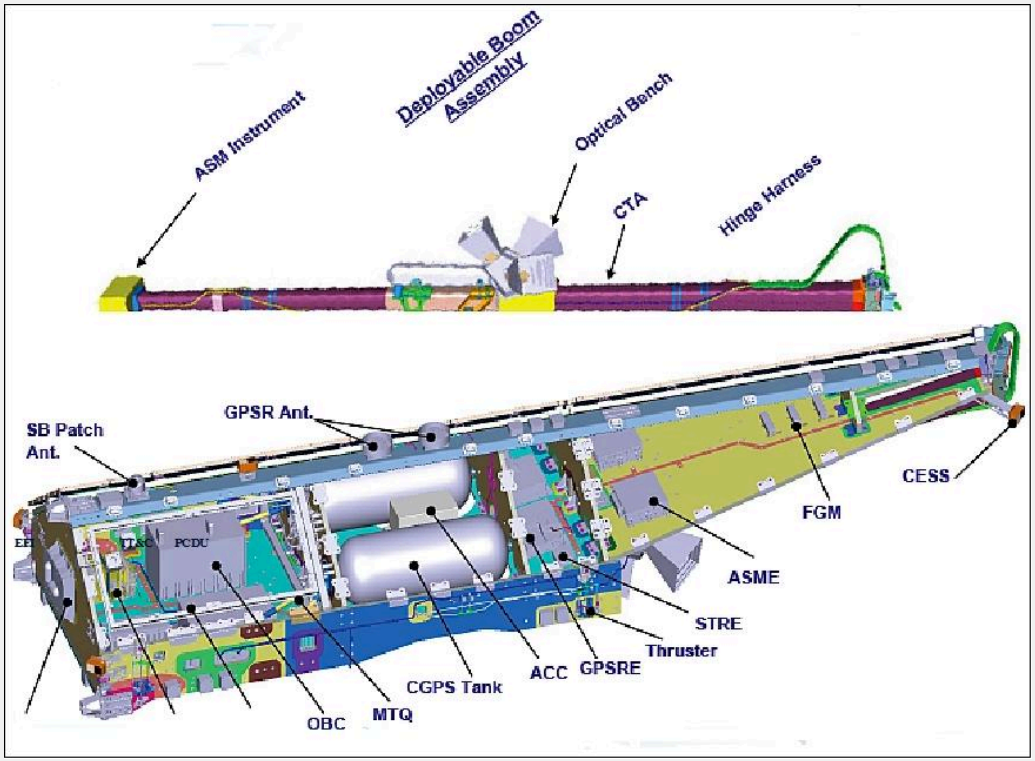






SWARM

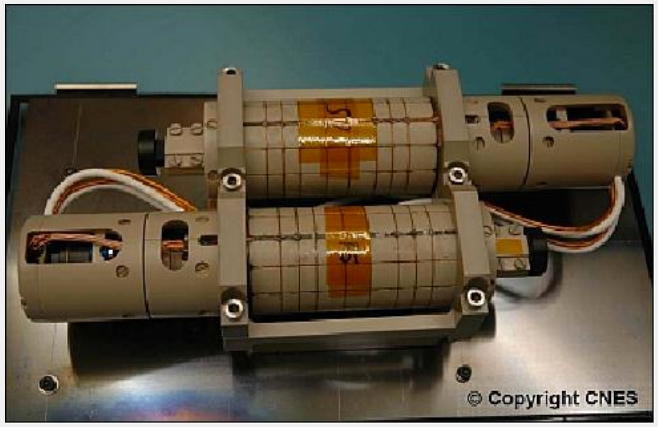
- Designed to replace Ørsted / CHAMP for magnetic field measurements
- Science objectives: Core dynamics, geodynamo processes, and core-mantle interaction; detailed model of field contributions
- Three spacecraft in two orbital planes
  - Two at 450 km; one at 530 km
- Each 9.1 x 1.5 x 0.85 meters; 5.1 meter boom
- Measures the B-field with < 1 nT accuracy and precision

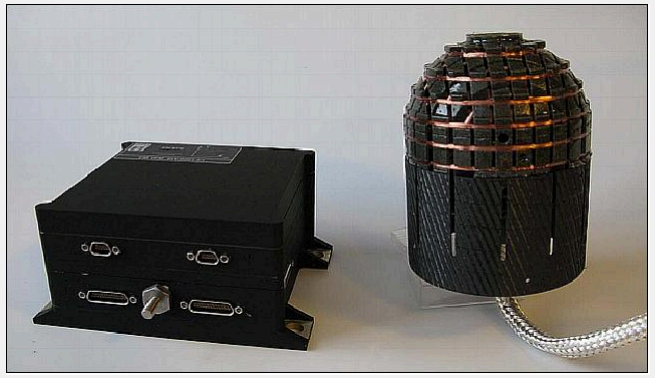


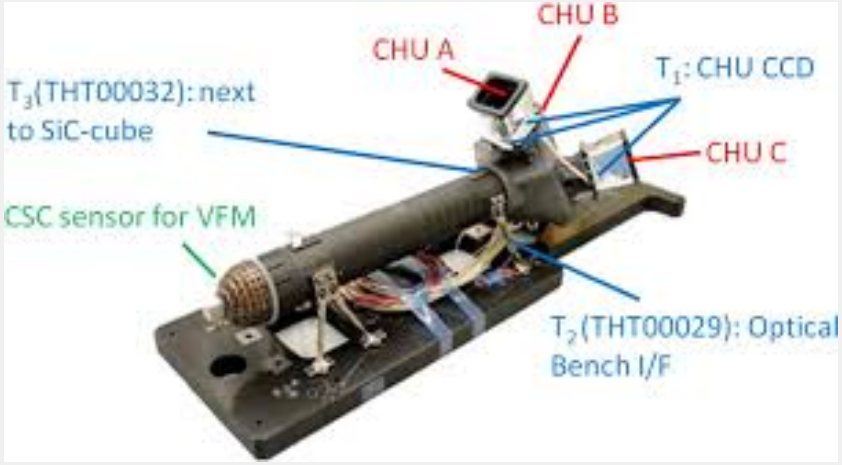
SWARM spacecraft: not small

What does it take to measure the B-field?

- Just a Magnetometer! Easy?
  - **Scalar magnetometer:** measures magnitude of the B-field, but not the vector direction
  - **Vector magnetometer:** measures the three components
    - Vector alone is not enough, VMs tend to have an offset (~100 nT) relative to absolute field magnitude
  - **Star Tracker:** need to know orientation of VM to 0.002 degree accuracy

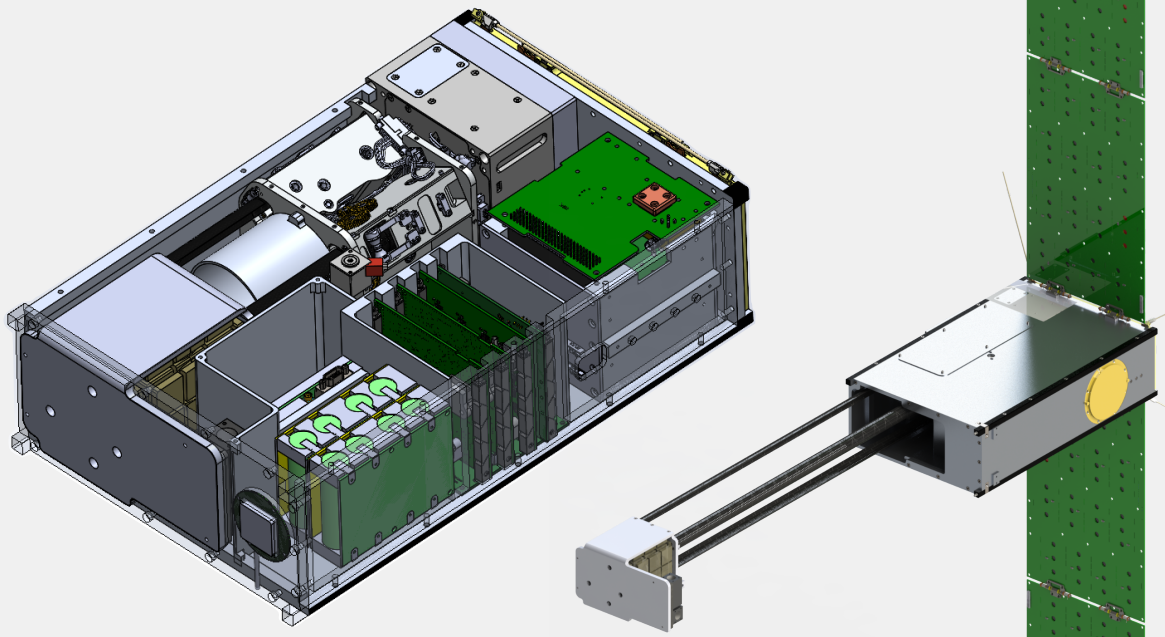


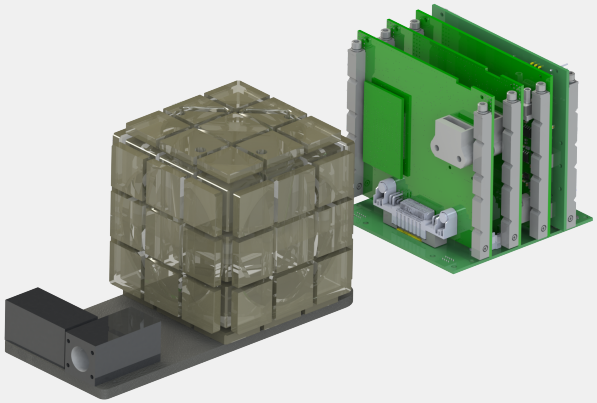


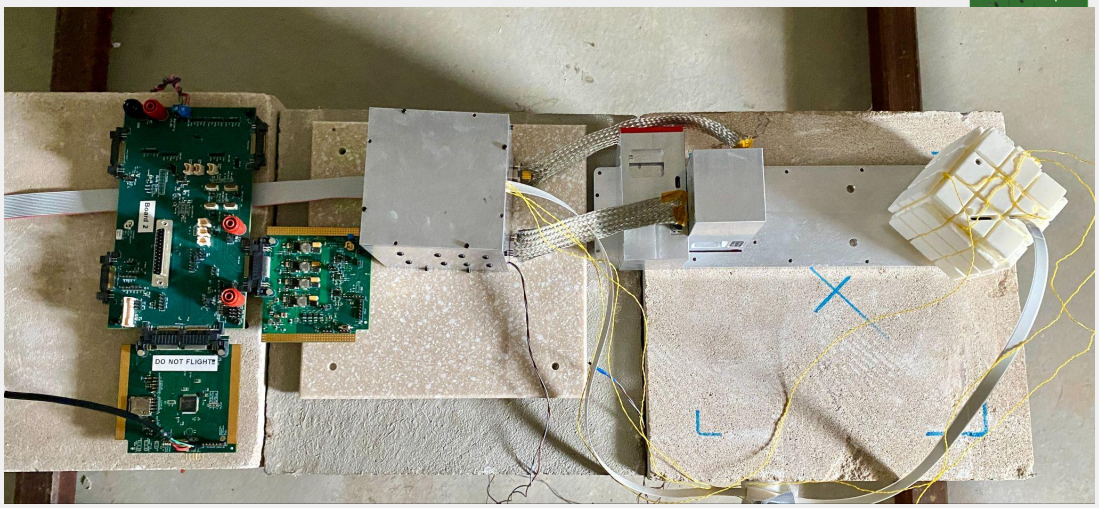


After SWARM: COSMO?

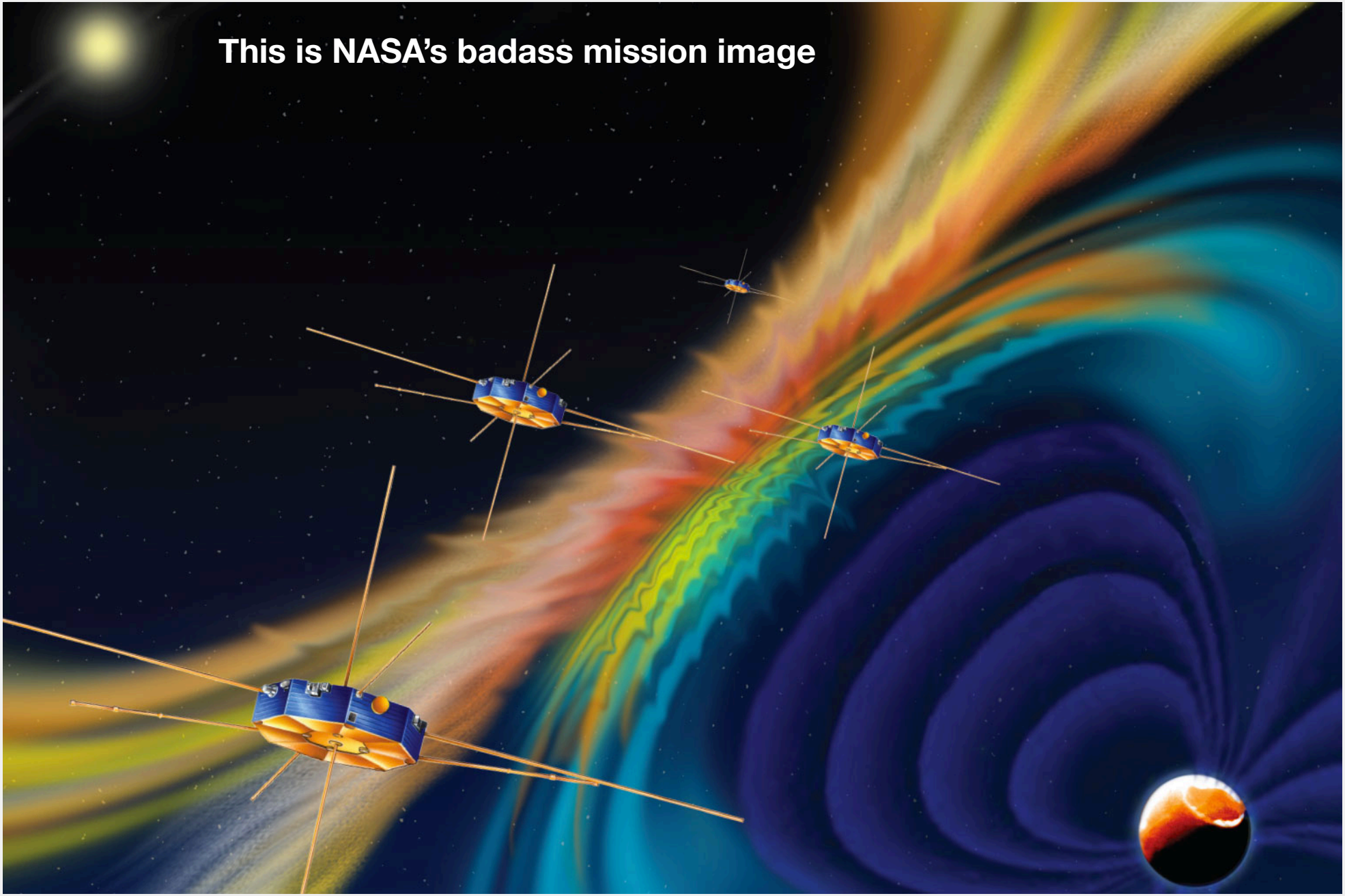
- **Can we do SWARM's job with a CubeSat?**
  - COSMO: Compact Spaceborne Magnetic Observatory
  - Using an ultra-compact Rubidium gas magnetometer (scalar)
    - Converting it into a scalar-vector magnetometer
  - COTS and custom spacecraft bus components
  - Designed and characterized for low noise, low bias fields



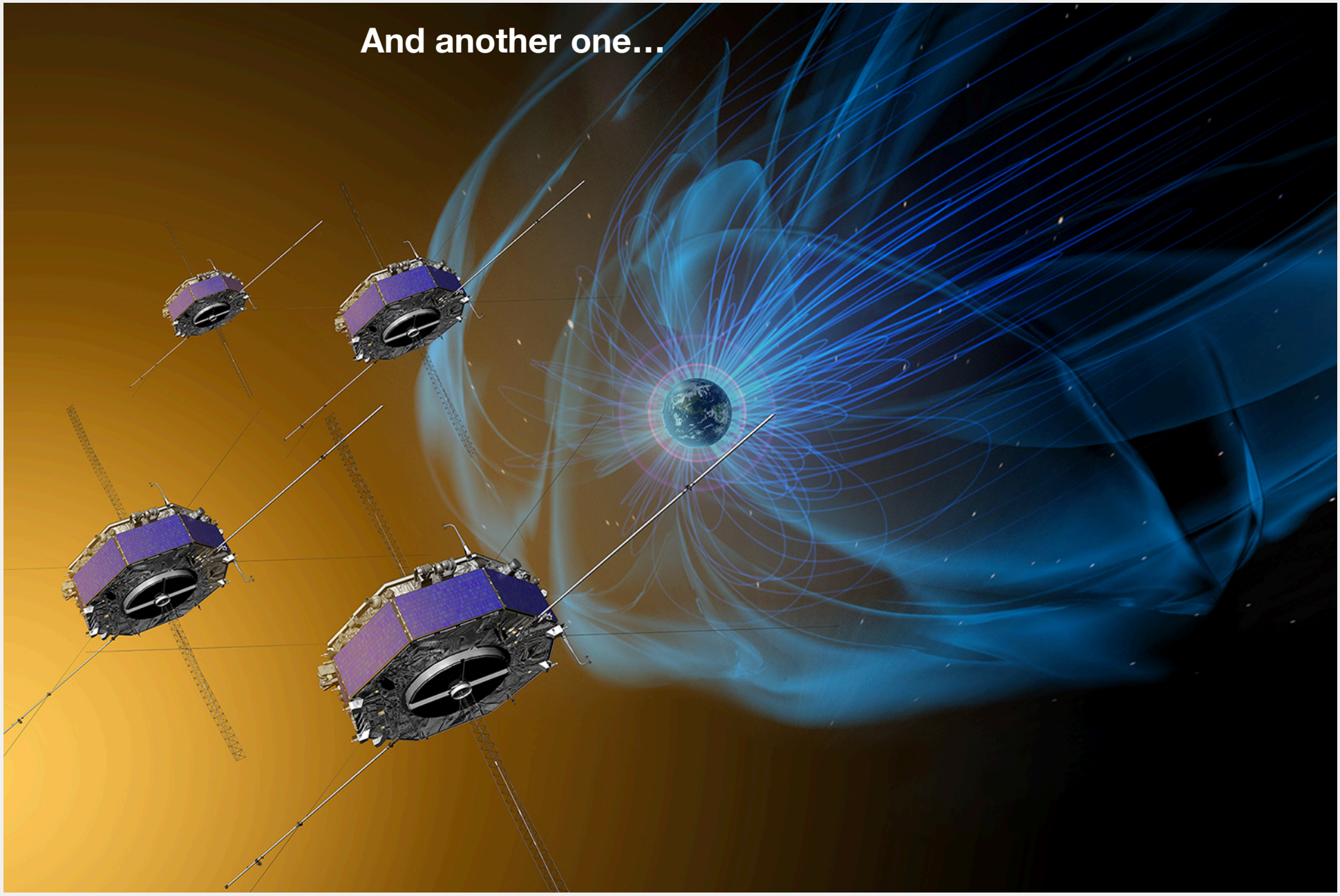




Magnetospheric Multiscale (MMS)



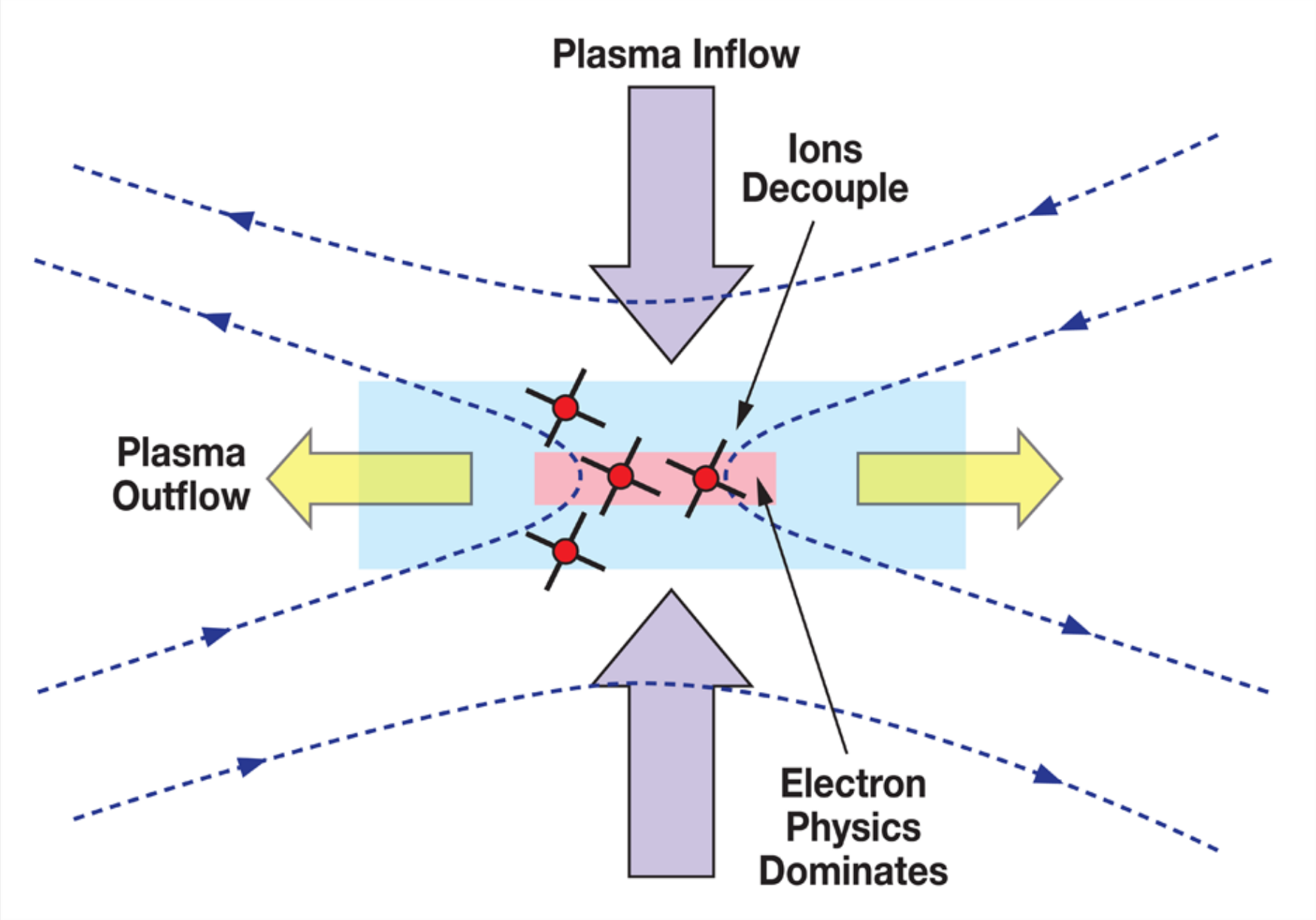
MMS



MMS Mission Goals

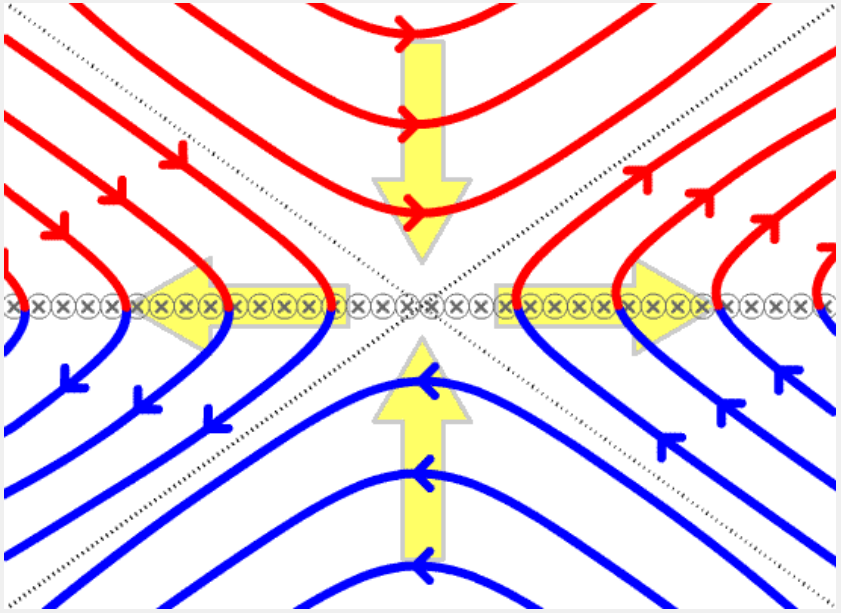
- **Why "multiscale"?**
  - Magnetosphere is enormous: 20 x 100 earth radii (R_E)
  - At reconnection, plasma physics is tiny: gyroscale
    (i.e., spatial scale limited by gyro radius: at B = 10 nT, r = 0.1–10 km
- **Understand process of reconnection on this tiny spatial scale**
  - Need a compact constellation of spacecraft: separation from 10-100's km
  - Complex attitude control / orbit design problem!
- **Scientific objectives:**
  - Determine the role played by electron inertial effects and turbulent dissipation in driving reconnection in the electron diffusion region
  - Determine the rate of magnetic reconnection and the parameters that control it
  - Determine the role played by ion inertial effects in the physics of reconnection

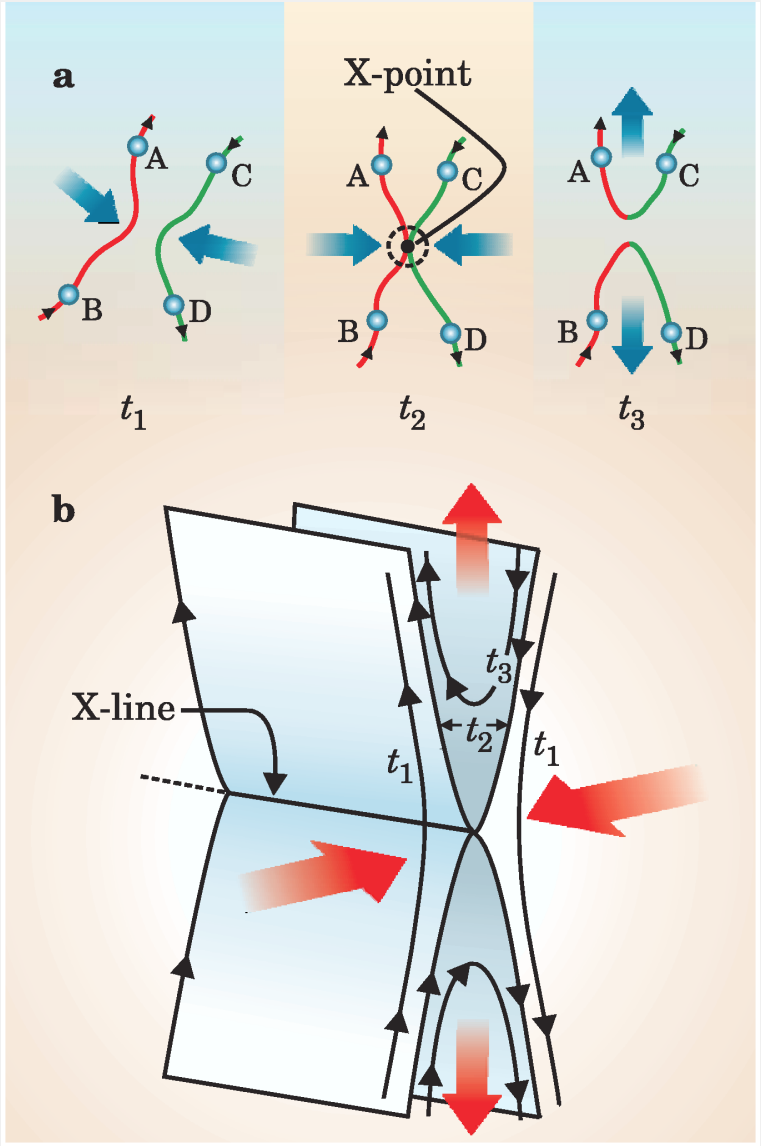
Reconnection



Reconnection

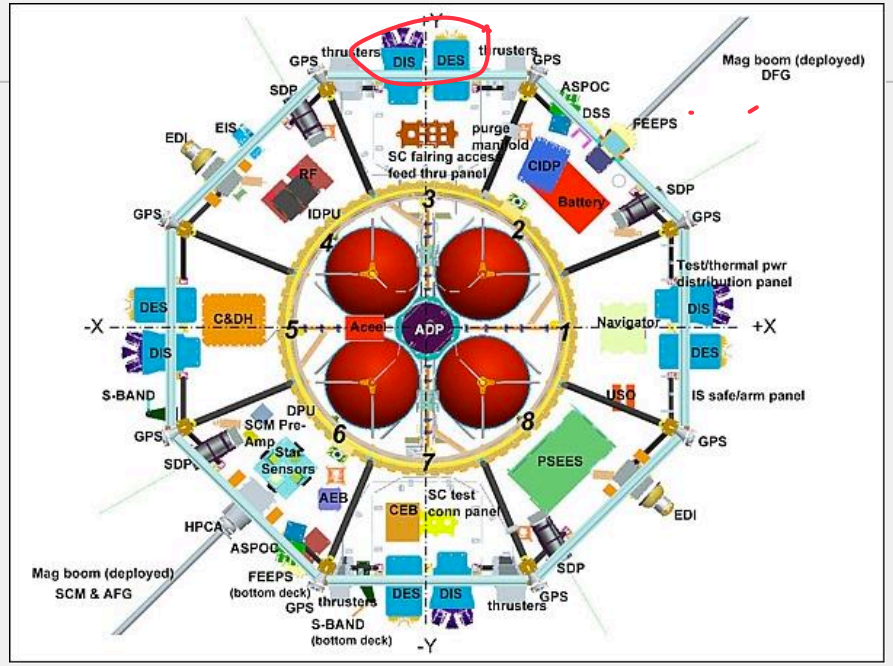
- Plasma flows towards X-line with parallel field lines
- After reconnection; plasma follows B-field in lateral direction
- Stored Magnetic energy is converted to kinetic energy of plasma; how much energy?

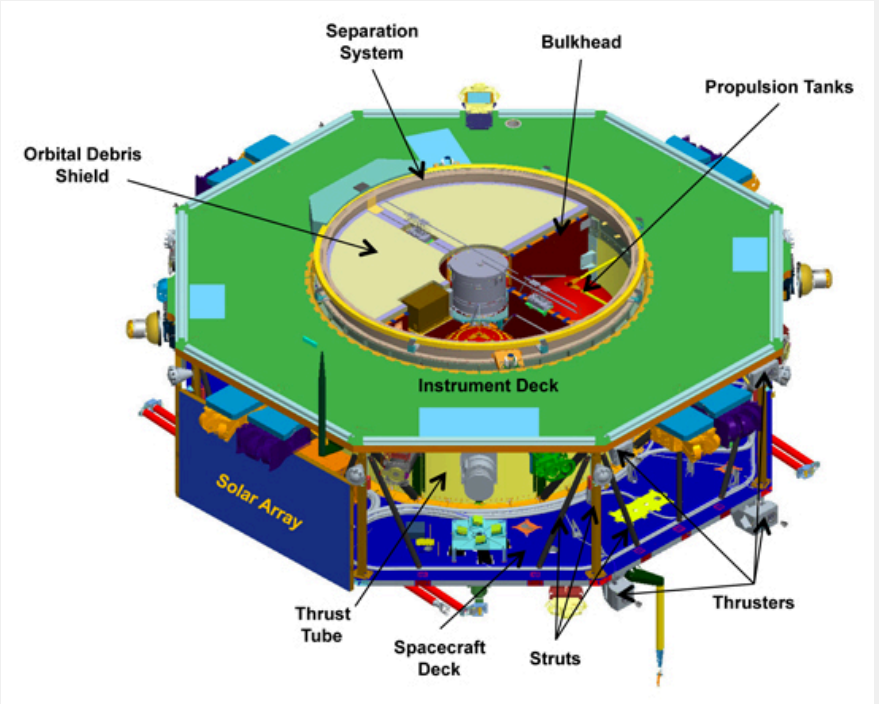




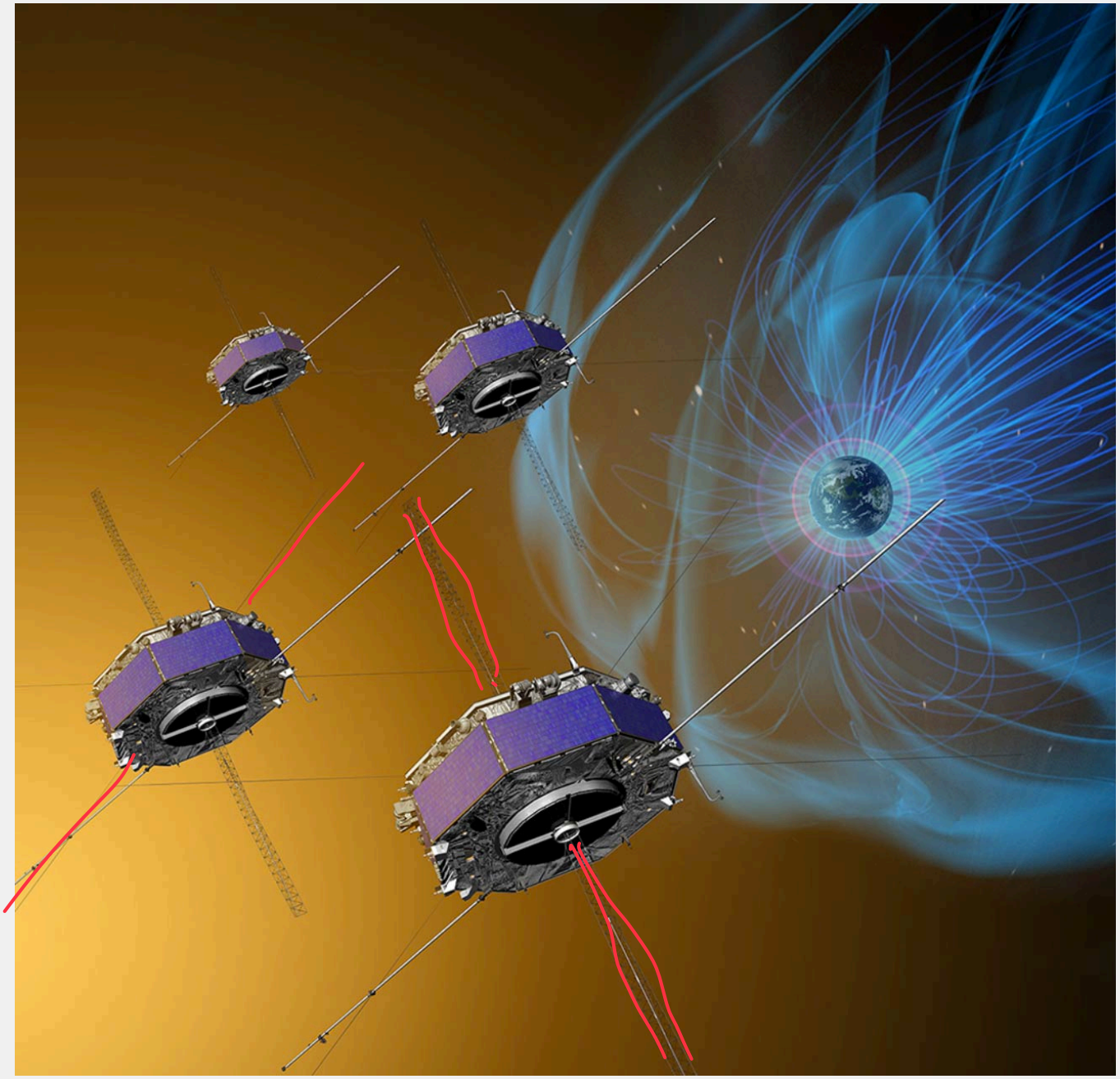
Instruments

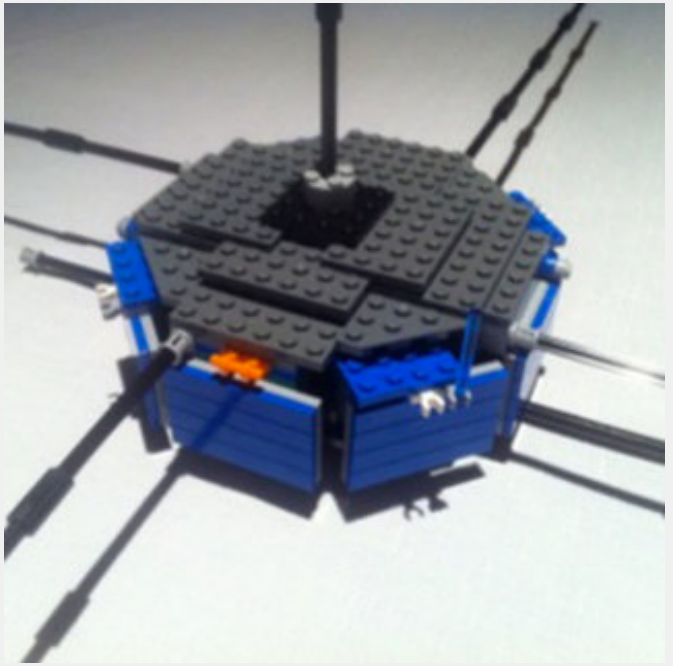
- **Hot Plasma composition suite**
  - Dual Ion sensors (DIS) and Dual electron sensors (DES); 10 eV – 30 keV
  - HPCA: oxygen and helium ions
- **Energetic Particles Detector Suite**
  - FEEPS: 25–500 keV electrons and ions
  - EIS: ion composition and angular distribution, 50–500 keV
- **FIELDS sensor suite**
  - AFG and DFG: Fluxgate magnetometers
  - EDI: electric and magnetic fields
  - Double probes: E-fields DC – 100 kHz
  - Search Coil magnetometers: B-fields DC – 6 kHz
- **Also:**
  - Inter-satellite communications
  - 360 kg on-board fuel for propulsion





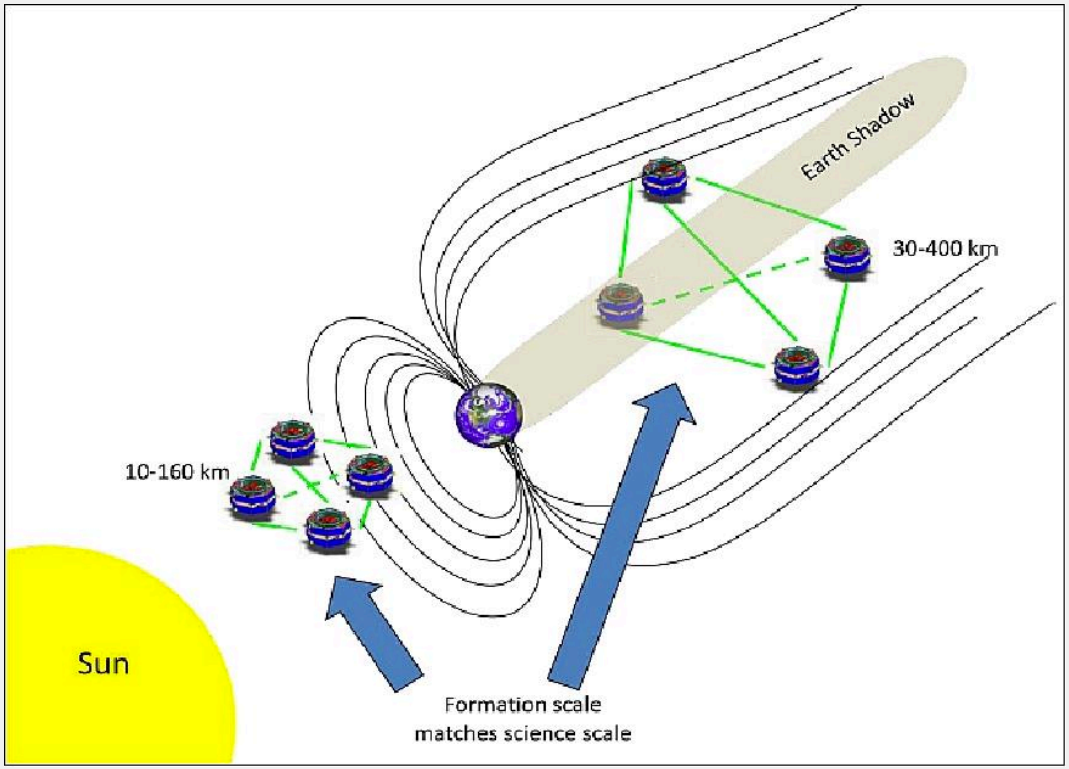
Deployed configuration





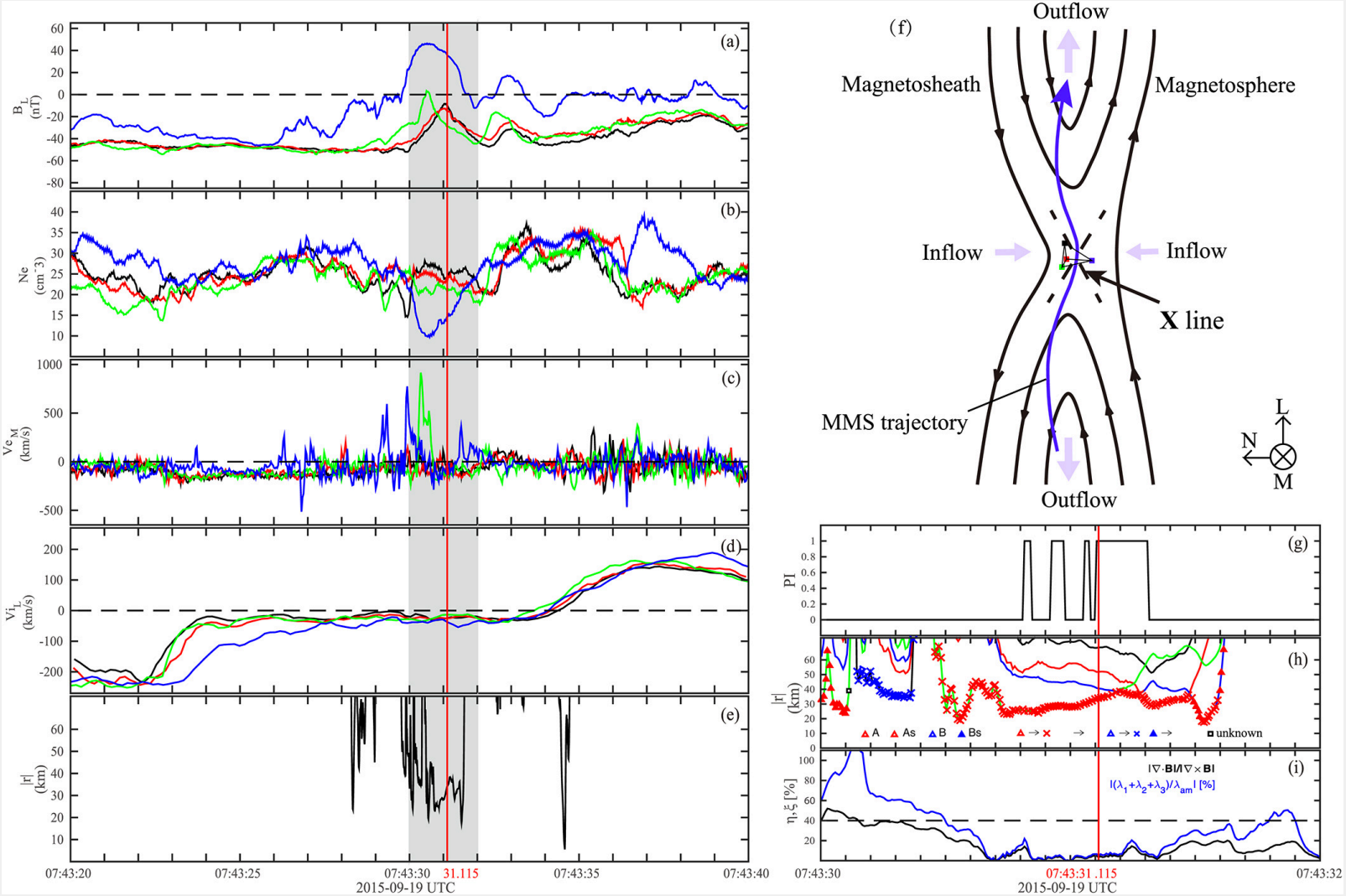
Launch and orbit configuration

- Four spacecraft had to be designed to “stack” together inside rocket body, and survive vibration
  - Vibe tested like this, seriously –>
- Deployable antennas stowed
- After deployment, move to tetrahedral formation



MMS in their own words

Reconnection Example... 🤔



Wang et al 2019, Figure 1: Overview of a reconnection event detected by MMS at the dayside magnetopause on 19 September 2015. Data are presented in the local magnetic normal (LMN) coordinates. Specifically, they are (a) the *L* component of magnetic field, (b) plasma density, (c) *M* component of electron bulk velocity, (d) *L* component of ion bulk velocity, and (e) the minimum distance between spacecraft and magnetic nulls resolved from the FOTE method. The vertical shade highlights an interval, during which magnetic null properties are analyzed in detail, including (g) the Poincare index, for distinguishing whether the null is enclosed by spacecraft tetrahedron, (h) the distance from magnetic null to each of the four MMS spacecraft, with the null types labeled, and (i) the two parameters, $\eta = |\nabla \cdot B|/|\nabla \times B|$ and $\xi = |\lambda 1 + \lambda 2 + \lambda 3|/|\lambda|\max$, for quantifying the uncertainty of the FOTE results. (f) A schematic illustrating the reconnection event and the trajectory of MMS. The red vertical line in (a)–(e) and (g)–(i) marks the time instant 07:43:31.115 UT, at which the electron distribution functions will be studied in detail.

The Core

The Core

- **Which of the following statements are correct?**
  - Everybody on Earth is dead in a year
  - Invisible field of energy made up of electricity and magnetism
  - Called the electromagnetic field
  - Protects us from cosmic radiation
  - The field is falling apart
  - Aaron Eckhart's description of the Earth's core
  - Physics 101, hot metal moving fast drives EM field
  - Planes will fall from the sky
  - Electronics will be fried
  - Static discharges in the atmosphere will lead to superstorms
  - EM field shields us from the solar winds, a lethal blend of radioactive particles and microwaves
  - Microwave radiation will literally cook our planet

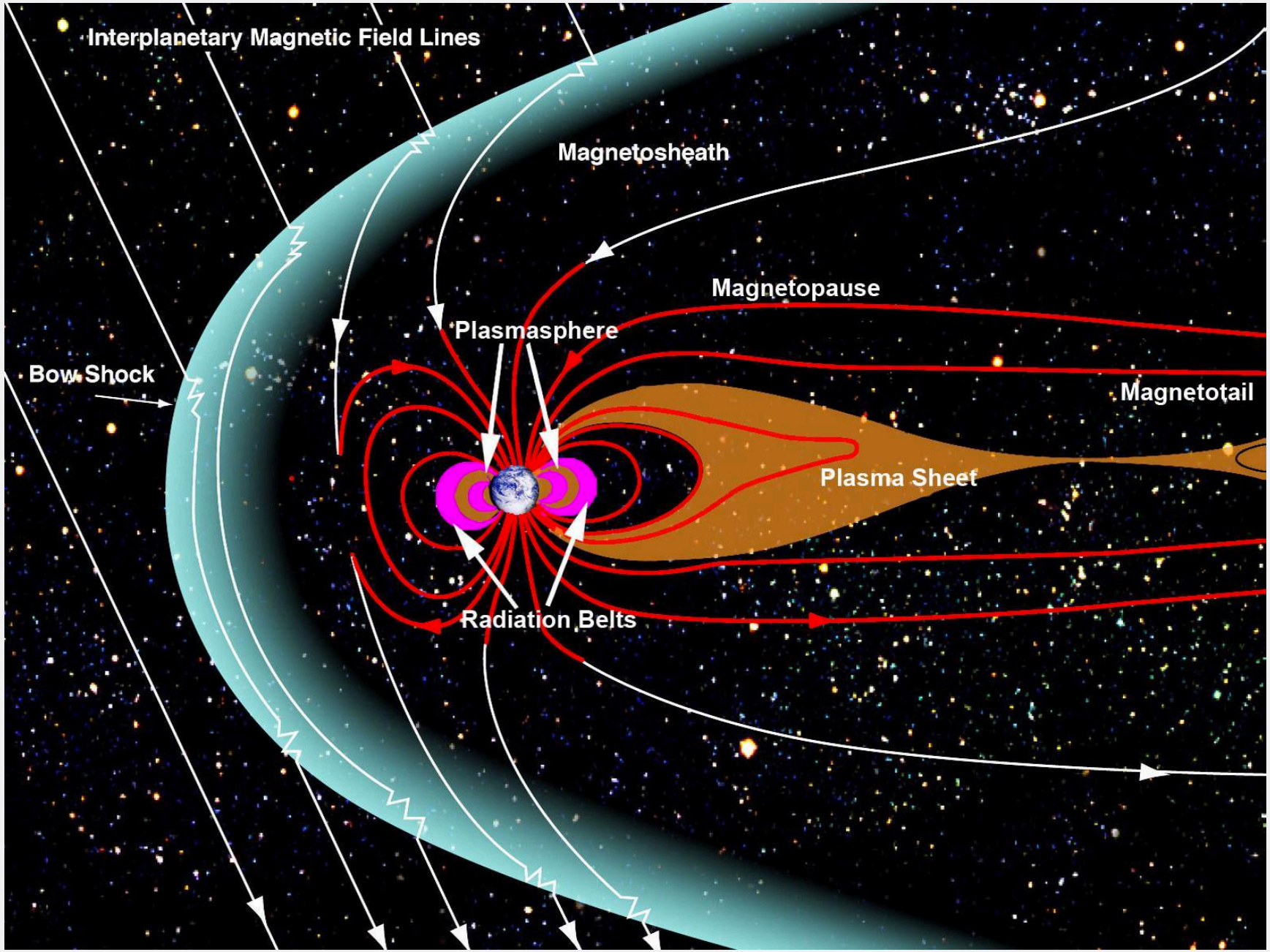
What really shields us from the solar wind etc?

- Solar wind itself is probably not that harmful;
  it requires > 100 MeV protons to reach the ground
- Solar SEPs, X-rays, gamma-rays are “radiation”
- There is nothing “radioactive”

- **What would happen if B-field disappeared?**

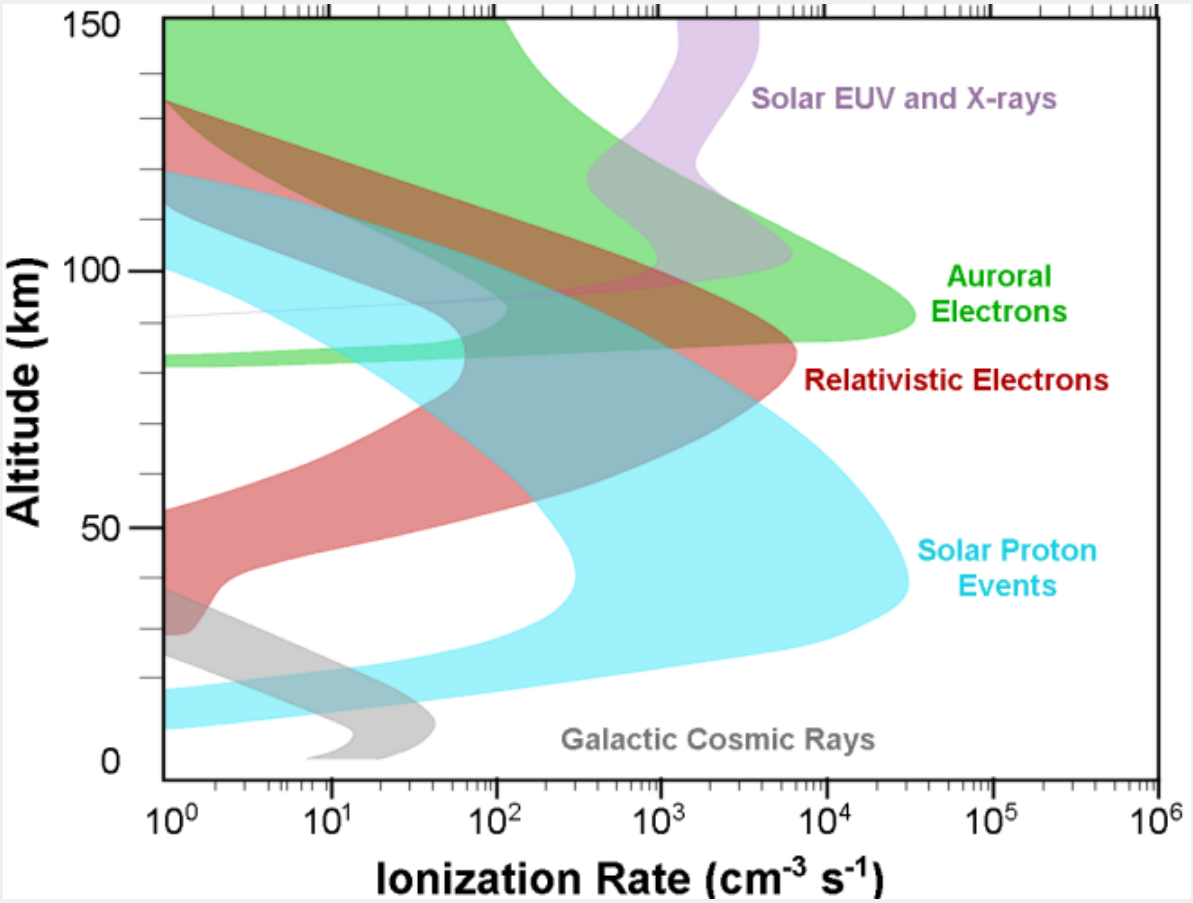
- **Where do X-rays, SEPs, etc get absorbed?**

How particles enter Magnetosphere

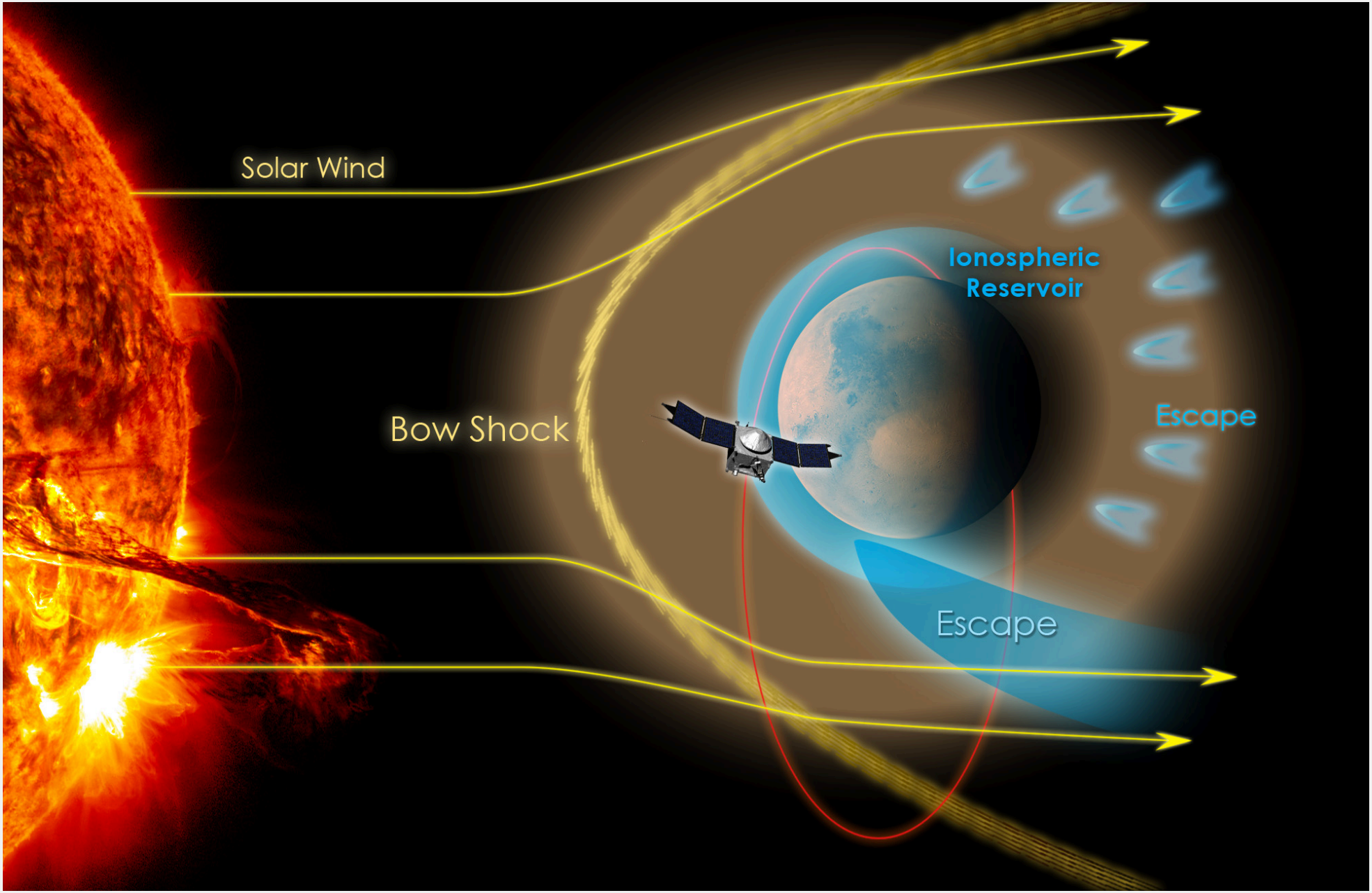


Where particles / X-rays get absorbed

- Energetic particles, EUV, X-rays, etc are absorbed by the atmosphere
- Only galactic cosmic rays reach the surface *(mostly secondaries);*
  - Earth's B-field actually does help shield us from them (about a factor of 2)
- SEPs only a concern near the poles, during events



If B disappeared…



Up Next

- **The Radiation Environment, including:**
  - **The (Van Allen) Radiation Belts**
  - probably the most important region from the perspective of damaging effects on spacecraft
  - Also the least understood.
  - **SEPs and GCRs too.**